

Stored Value Support Manual

BASE24-pos[®]



© 2006 by ACI Worldwide Inc. All rights reserved.

All information contained in this documentation is confidential and proprietary to ACI Worldwide Inc., and has been delivered for use pursuant to license. No part of this documentation may be photocopied, electronically transferred, modified, or reproduced in any manner that is contrary to license without the prior written consent of ACI Worldwide Inc.

Access Control Server, ACI Automated Case Management System, ACI Card Management System, ACI Claims Manager, ACI Commerce Gateway, ACI Debit Card System, ACI e-Courier, ACI Host Link, ACI Merchant Accounting System, ACI Payments Manager, ACI Proactive Risk Manager, ACI Smart Chip Manager, BASE24, BASE24-atm, BASE24-card, BASE24-check auth, BASE24-configuration manager, BASE24-es, BASE24-frequent shopper, BASE24-infobase, BASE24-pos, BASE24-refunds, BASE24-remote banking, BASE24-teller, NET24, and WINPAY24 are trademarks or registered trademarks of ACI Worldwide Inc., Transaction Systems Architects, Inc., or their subsidiaries.

Other companies' trademarks, service marks, or registered trademarks and service marks are trademarks, service marks, or registered trademarks and service marks of their respective companies.

Contents

Preface	vii
Conventions Used in This Manual	xi
1: Introduction	1-1
Stored Value Support	1-2
Stored Value Transaction Types	1-2
Default Account Type	1-3
Stored Value Card Types	1-4
LCONF Assigns and Params	1-5
Assigns	1-5
Params	1-6
Stored Value Token	1-8
2: Files Maintenance	2-1
ACI Standard Device Configuration File (ACNF)	2-2
ACNF Screen 8	2-2
ACI Standard Device Response File (ARSP)	2-4
ARSP Screen 5	2-4
Card Prefix File (CPF)	2-6
CPF Screen 6	2-7
CPF Screen 7	2-8
CPF Screen 10	2-9
Cardholder Authorization File (CAF)	2-13
CAF Screen 10	2-13
Extract Configuration File (ECF)	2-14
ECF Screen 7	2-14
Extract Screen (EXTR)	2-15
Extract Screen 7	2-15

Institution Definition File (IDF)	2-16
IDF Screen 1	2-16
POS Transaction Log File (PTLF)	2-17
PTLF Screen 4	2-17
Positive Balance File (PBF)	2-19
Stored Value History File (SVHF).....	2-20
Key Screen Function Keys	2-22
SVHF Key Perusal Selection Screen 1	2-23
SVHF Card Summary List Screen 2.....	2-31
SVHF Detail Screen 3 — Format 1	2-34
SVHF Transaction Code Summary List Screen 2	2-37
SVHF Detail Screen 3 —Format 2.....	2-40
SVHF Retailer Summary List Screen 2	2-41
SVHF Detail Screen 3—Format 3	2-44
SVHF Retailer and Terminal Summary List Screen 2	2-45
SVHF Detail Screen 3—Format 4	2-48
Stored Value Dormancy File (SVDF).....	2-49
 3: Authorization Processing	 3-1
CAF and PBF Usages	3-2
Adding CAF Records	3-3
Card Activation Transaction	3-6
Additional Card Activation Transaction.....	3-9
Replenishment Transaction.....	3-10
Full Redemption Transaction	3-12
Offline Stored Value Transactions	3-13
 4: SPDH Message Formats	 4-1
Transaction Codes.....	4-2
Terminal Messages	4-3
Card Activation Transaction.....	4-3
Additional Card Activation Transaction.....	4-3
Replenishment Transaction.....	4-3
Full Redemption Transaction	4-4
Card Activation Transaction Request	4-5

Card Activation Transaction Response	4-7
Additional Card Activation Transaction Request	4-9
Additional Card Activation Transaction Response	4-11
Replenishment Transaction Request	4-12
Replenishment Transaction Response	4-14
Full Redemption Transaction Request	4-16
Full Redemption Transaction Response	4-18
5: ISO Host Message Formats	5-1
Transaction Codes	5-2
Terminal Messages	5-3
Card Activation Transaction	5-3
Additional Card Activation Transaction	5-3
Replenishment Transaction	5-3
Full Redemption Transaction	5-4
6: Batch Processing and Reporting	6-1
History File Cleanup Module (HFCLUP)	6-2
Program Startup	6-2
Reports	6-3
HFCLUP Processing Steps	6-4
Stored Value History File Extract Module (SVHFX)	6-6
CAF/PBF Cleanup Program (CARDCLN)	6-7
Process Startup	6-7
Reports	6-8
CAF/PBF Cleanup (CARDCLN) Processing Steps	6-9
BASE24-pos Reports	6-10
7: Dormancy and Escheatment	7-1
Dormancy Processing and Escheatment Reporting	7-2
Initialization Steps	7-2
Account Processing	7-5
Dormancy Report	7-12
Escheatment Report	7-13

Index Index-1

Index by Field Name. Index-3

Index by Data Name Index-5

Preface

The *BASE24-pos Stored Value Support Manual* discusses the four Stored Value transactions in detail. It also provides the configuration changes to the various BASE24 files and modules required for the Stored Value add-on product.

Audience

This manual is intended for personnel who must understand the implementation of the BASE24-pos Stored Value add-on product.

Prerequisites

This manual assumes the reader is familiar with the BASE24-pos product. Only the additions and changes that have been made for the Stored Value module are described in the manual. Where appropriate, references for more information are made to BASE24-pos manuals.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need to know the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The *BASE24 Base Files Maintenance Manual* provides a comprehensive explanation of the BASE24 screens that access the base and switch files.
- The *BASE24 Logical Network Configuration File Manual* describes the BASE24 assigns and params.

-
- The ***BASE24 Tokens Manual*** contains specifications for tokens and token processing.
 - The ***BASE24-pos Files Maintenance Manual*** provides descriptions of the BASE24 screens that access BASE24-pos files and functions.
 - The ***BASE24-pos Settlement and Reporting Manual*** describes the settlement and reporting tools available with the BASE24-pos product. It also describes the functionality within the BASE24-pos product that guides and controls cutover and reporting actions.
 - The ***BASE24-pos Standard POS Device Configuration Manual*** provides detailed information on configuring the BASE24 Standard POS Device Handler (SPDH) module and the files used with this module.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an introduction to the functionality, benefits, and the transactions that are supported by the BASE24-pos Stored Value product.

Section 2, “Files Maintenance,” provides details of BASE24 screens that support the BASE24-pos Stored Value transactions.

Section 3, “Authorization Processing,” describes additional authorization processing that is performed on Stored Value transactions.

Section 4, “SPDH Message Formats,” describes the SPDH message formats that are specific to Stored Value processing.

Section 5, “Batch Processing and Reports,” describes those batch processes used by the Stored Value product. It also provides the BASE24-pos reports that contain Stored Value data.

Section 6, “Dormancy and Escheatment,” describes the processing that is common to both dormancy processing and escheatment reporting and includes examples of dormancy and escheatment reports.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, SV-AE000-01 for the ***BASE24-pos Stored Value Support Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Nov-2006 for November, 2006) indicates the issue of the manual. The software release information (for example, R6.0v7 for release 6.0, version 7) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

ACI Worldwide Inc.

Conventions Used in This Manual

This section explains how field descriptions are documented in this manual.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from Card Prefix File (CPF).

Example

TOTAL PER REFUND/REPLENISH — The maximum amount allowed for a single/refund replenishment against credit and noncredit accounts using BASE24-pos and the Stored Value add-on product.

Field Length: 15 numeric characters depending upon currency being used
Required Field: Yes
Default Value: 0
Data Name: CPF.POSCPF.TTL-CR-PER-RFND-LMT

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is <i>System Protected</i>.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol *b* to indicate the blank character.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

ACI Worldwide Inc.

1: Introduction

The BASE24-pos Stored Value add-on product allows retailers and processors to issue a stored value card. All account information is stored on BASE24 as it is the database of record. Retailers and or processors are able to establish minimum and maximum replenishment amounts and configure the maximum amount of cash that can be returned to the customer.

The stored value card allows retailers to replace paper gift certificates, scrip, merchandise vouchers, and in-store vouchers with stored value cards. The concept can also be used for business incentive programs.

Stored Value Support

This document discusses the four Stored Value transactions in detail. It also provides the configuration changes to the various BASE24 files and modules required for the Stored Value add-on product.

Some of the features of the Stored Value add-on product are:

- Flexible expiration date configuration methods
- Transaction history
- Card blocking and replacement
- Minimum and maximum activation and replenishment amounts
- Card activation
- Additional card activation
- Replenishment of funds
- Remaining card balance
- Cash back
- Maximum card balance
- Multiple program support

Stored Value Transaction Types

A retailer or processor location is assigned specific BINs for issuing of stored value cards. The customer then purchases a stored value card from the retailer. The clerk initiates a card activation transaction online using a modified POS device. This transaction causes BASE24 to update the card status for the card to be issued in the Cardholder Authorization File (CAF).

Once a card is activated, cardholders can initiate any standard POS transaction supported by the BASE24 Standard POS Device Handler (SPDH) module provided the retailer's or processor's terminals support those transactions. For more information about the SPDH, refer to the ***BASE24-pos Standard POS Device Configuration Manual***.

The SPDH communicates with the modified POS device. It allows the retailer and or processor to support all standard POS transactions as well as the following Stored Value transactions:

Card Activation transaction — A card activation transaction is initiated when a customer swipes the card at the POS device. That card's PAN and an amount to be loaded (if it isn't preloaded) in the account, are provided to BASE24 and used to activate a CAF record and corresponding Positive Balance File (PBF) record. If a CAF record does not exist, a default CAF record is read and used as a CAF template record. This template record is then used to add the new CAF record.

Note: For cards that are preloaded in the database, the Primary CAF status must be 0 (Issued not active for this transaction type).

Additional Card Activation transaction — The additional card activation transaction allows the customer to obtain additional cards that access the same account as their original card. To activate additional cards, the customer swipes an existing card and an additional card at the POS device. PBF balances are not impacted by an additional card activation transaction.

Replenishment transaction — A replenishment transaction is initiated when the customer requests additional funds to be loaded to an existing value card. The "maximum card balance value" restricts the amount that can be placed into an account.

Note: This transaction can only be accomplished with cards defined as reloadable.

Full Redemption transaction — A full redemption transaction subtracts the remaining funds from an account. The card and account status are not altered. Redemption transactions are only allowed if there are no pre-authorization holds present and the account balance is not greater than the "maximum balance as cash" limit specified in the CPF.

Note: Stored Value transactions are authorized using the offline-only authorization level and the CAF/PBF authorization method.

Default Account Type

Because Stored Value transactions may include an invalid account type or no account type at all, the Router/Authorization module defaults the account type to 01 (DDA for all transactions).

Stored Value Card Types

The BASE24-pos stored value add-on product supports the following user-defined card types:

- Stored Value no Reload (SN)
- Stored Value Reload (SR)

These card types are configured in the COBNAMES file (CARD-TYPE-TABLE).

LCONF Assigns and Params

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for BASE24. There is one LCONF for each logical network.

The LCONF contains *assigns* that link the physical names of objects in the logical network to names used in the BASE24 code, and *params* that supply parameters and processing values to BASE24.

The BASE24-pos Stored Value add-on product modules use the standard LCONF assigns and params, together with the addition of the assigns and params included in the descriptions in this section.

For details of the LCONF assigns and params, including those used by the BASE24-pos Stored Value product, refer to the ***BASE24 Logical Network Configuration file Manual***.

Assigns

The following assigns are used by the BASE24-pos Stored Value add-on product.

- The POS-SV-PTLF-EXTMEM-SWAPVOL assign specifies the fully qualified name of the BASE24-pos Stored Value Dormancy/Escheatment extended memory swap volume. This process uses this assign to temporarily store data from the PTLF when building extended memory. If this assign is not present in the LCONF or contains an invalid value, the Stored Value Dormancy/Escheatment program uses the volume in which it resides.
- The POS-SVHF assign, which is used by the Router/Authorization module, identifies the name of the Stored Value History File (SVHF). Transactions using a Stored Value card are logged to this file when the BASE24-pos Stored Value add-on product is installed. If transactions are routed to an interchange for authorization, logging to this file is optional.
- The PRI-COL assign specifies the fully qualified PPD name of the primary EMS Collector process.
- The ALT-COL assign specifies the fully qualified PPD name of the alternate EMS Collector process.
- The CPF assign specifies the fully qualified file location of the Card Prefix File.

- The POS-SV-CPF-EXTMEM-SWAPVOL assign specifies the fully qualified name of the BASE24-pos Stored Value CPF extended memory swap volume. This process uses this assign to temporarily store data when building extended memory. If this assign is not present in the LCONF or contains an invalid value, the NSK operating system assigns a swap volume location.
- The IDF assign specifies the fully qualified file location of the Institution Definition File.
- The POS-SV-IDF-EXTMEM-SWAPVOL assign specifies the fully qualified name of the BASE24-pos Stored Value IDF extended memory swap volume. This process uses this assign to temporarily store data when building extended memory. If this assign is not present in the LCONF or contains an invalid value, the NSK operating system assigns a swap volume location.
- The POS-SVDF assign specifies the file location of the Stored Value Dormancy File.
- The POS-SV-SVDF-EXTMEM-SWAPVOL assign specifies the fully qualified name of the BASE24-pos Stored Value SVDF extended memory swap volume. This process uses this assign to temporarily store data when building extended memory. If this assign is not present in the LCONF or contains an invalid value, the NSK operating system assigns a swap volume location.
- The POS-SV-DORMANCY-RPT assign specifies the print location for the dormancy report.
- The POS-SV-ESCHEATMENT-RPT assign specifies the print location for the escheatment report.
- The POS-SVHF assign specifies the file location of the Stored Value History File.

Params

The following params are used by the BASE24-pos Stored Value add-on product.

- The POS-SV-PTLF-LOGGING param specifies whether Stored Value transactions should be logged to the POS Transaction Log File (PTLF).
- The POS-SV-RETAILER-ID param specifies the POS retailer identification number. This param is used by the Dormancy or Escheatment process when logging transactions to the PTLF.

- The POS-SV-TERMINAL-ID param specifies the POS terminal identification number. This param is used by the Dormancy or Escheatment process when logging transactions to the PTLF.
- The POS-SVHF-ICHG-AUTH-LOG param specifies whether the interchange authorized transactions should be logged to the Stored Value History File (SVHF).
- The POS-SVHF-NOTIFY-INTERVAL param defines the number of writes made to the SVHF between calculating the percentage full of the SVHF.
- The POS-SVHF-NOTIFY-PERCENTAGE param specifies the percentage of file full at which the Router module of the BASE24-pos Device Handler/ Router/Authorization process generates an event message indicating that the Stored Value History File (SVHF) is becoming full.
- The POS-SVHF-RET-PERIOD param specifies the number of days worth of history data to retain in the Stored Value History File (SVHF) when the History File Cleanup module (HFCLUP) is run. This param is required when the BASE24-pos Stored Value add-on product is installed and the SVHF is used. The History File Cleanup module (HFCLUP) uses this param.
- The POS-SVHF-NOTIFY INTERVAL param specifies the number of writes made to the SVHF between calculating the percentage full of the SVHF.
- The POS-SVHF-NOTIFY PERCENTAGE param specifies when to log a message based on the percentage full of the SVHF.

Stored Value Token

The Router/Authorization module looks for the Stored Value token (U2) on all transactions initiated with a Stored Value card. If the transaction is a Stored Value transaction, the Router/Authorization module adds the Stored Value token if it is not found. The Router/Authorization module then updates the token with the following information:

- PBF balance
- CAF expiration date
- CPF maximum balance as cash flag
- Reversal flag

The available balance in the Stored Value token is taken from the available balance from the PBF minus the total pre-auth holds. The balance can be negative if the purchase amount is greater than the available balance minus the total pre-authorization holds, in which case the negative amount returned will indicate that portion of the purchase amount that must be tendered through some other form of payment.

For more information about BASE24 tokens, including a description of the Stored Value token, see the ***BASE24 Tokens Manual***.

2: Files Maintenance

This section provides details of BASE24 screens that support the BASE24-pos Stored Value transactions. For information about other BASE24-pos screens, see the *BASE24 Base Files Maintenance Manual* and the *BASE24-pos Files Maintenance Manual*.

For new records, the Stored Value request transactions should have the FIDs required by the SPDH set to a value of Y. Please refer to the table below for the required FIDs. Like standard processing, the message authentication FIDs all default to a value of N.

Stored Value Transactions	Request		Response	
	Required FIDs	Required SFIDs	Required FIDs	Required SFIDs
Card Activation	B, q		B, q, 6	S
Additional Card Activation	B, q, 6	S	q, 6	S
Replenishment	B, q		B, q, 6	S
Full Redemption	B, q		q, 6	S

ACI Standard Device Response File (ARSP)

The ACI Standard Device Response File (ARSP) allows customers to alter the responses sent to the terminal, depending on the transaction being sent and the language being used. For further explanation of the ARSP, refer to the *BASE24-pos Standard POS Device (SPDH) Manual*.

ARSP Screen 5

ACI Standard Device Response File (ARSP) screen 5 contains the Stored Value description fields required to support the stored value transactions.

Note: An ARSP record must be created even though no response alteration is required.

A sample ARSP screen 5 is shown below, followed by descriptions of its fields.

```
BASE24-pos   SPDH ARSP MAINT           LLLL           YY/MM/DD  HH:MM  05 OF 05
              TERMINAL GROUP:           RECORD NUMBER:  04

              DESCRIPTION RECORD

<original description>  <customer description>
SHIFT SUBTOT REQUEST
DAY SUBTOT REQUEST
READ MAIL REQUEST
MAIL DELIVERED REQUEST
SEND MAIL REQUEST
DOWNLOAD REQUEST
HANDSHAKE REQUEST
RESERVED
CARD ACTIVATION
ADNL CARD ACTIVATE
REPLENISHMENT
FULL REDEMPTION

***** BASE24 *****
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
                  F12-HELP
```

For new records, the descriptions default to spaces like the standard transaction descriptions.

DESCRIPTION RECORD

The following fields describe supported transactions with original descriptions and customer descriptions.

<original description> — The transaction description as it is known by the SPDH module.

Field Length:	System protected
Occurs:	Up to 12 times per screen

<customer description> — The transaction description as defined by the customer for the purpose of responses.

Field Length:	1 to 24 alphanumeric characters
Occurs:	Up to 12 times per screen
Required Field	No
Default Value:	No default value
Data Name:	ARSP.DESCR.TBL

Card Prefix File (CPF)

The Card Prefix File (CPF) defines each card prefix that can be processed within a BASE24 logical network. One record must exist in the CPF for each card prefix to be processed. If one prefix is used with multiple primary account number (PAN) lengths, multiple records must exist in the CPF—one for each PAN length.

CPF records define the characteristics of each prefix and also contain prefix-specific parameters that allow institutions to define portions of their authorization processing that can be controlled at the prefix level. These parameters include expiration date checks, card and PIN verification controls, withdrawal limits, and credit account minimum standard increments.

The **EXP CHECK TYPE** field on screen 1 of the CPF indicates the type of card expiration date check to use for this prefix. Valid values are as follows:

- 0 = Do not check the expiration date.
- 1 = Check the expiration date on Track 1 or Track 2.
- 2 = Check the expiration date in the CAF record.

CPF Screen 6

CPF screen 6 contains BASE24-pos authorization information, including activity limits for BASE24-pos.

A sample CPF screen 6 is shown below, with the fields relevant to configuration of the BASE24-pos Stored Value add-on product highlighted. Refer to the ***BASE24 Base Files Maintenance Manual*** for descriptions of these fields.

BASE24-pos	CARD PREFIX	LLLL	YY/MM/DD	HH:MM	06 OF 22
PREFIX: 1111	PAN	LENGTH: 14	FIID:		BK01

POS AUTHORIZATION INFORMATION

ACTIVITY LIMITS:	TOTAL	OFFLINE
CASH WDL:	100000	100000
CASH ADV:	100000	100000
AGGR:	100000	100000
PURCHASE:	100000	100000
PER REFUND/REPLENISH:	999	999
REFUND/REPLENISH:	999	999

MINIMUM CASH ADV AMOUNT: 0

STANDARD CASH ADV INCREMENT: 0

TIMES USED PER PERIOD LIMIT: 1

ISSUER TXN PROFILE:

***** BASE24 *****

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

CPF Screen 7

CPF screen 7 contains an activity limit field for refund and replenishment transactions. A sample of CPF screen 7 is shown below, with this field highlighted. Refer to the ***BASE24 Base Files Maintenance Manual*** for a detailed description of this field.

```
BASE24-pos      CARD PREFIX      LLLL      YY/MM/DD  HH:MM  07 OF 22
PREFIX: 1111      PAN LENGTH: 14      FIID: BK01

      POS INFORMATION

MAXIMUM NUMBER OF REFUND/REPLENISH: 9999
      MAXIMUM NUMBER OF PRE-AUTH HOLDS: 0
      CHARGEBACK UPDATE: N (DO NOT UPDATE DATABASE)
      REPRESENTMENT UPDATE: N (DO NOT UPDATE DATABASE)
      DEFAULT COMBO CARD TYPE: 2 (DEBIT CARD)
      DEFAULT ACCOUNT TYPE: 01 (CHECKING)
                           31 (CREDIT ACCT)
                           11 (SAVINGS)
      POS PROCESSING INFORMATION

      ISSUER: 00 ON-US
      PIN PROCESSING FLAG: 0 (PIN NOT REQUIRED)
      ADDRESS VERIFICATION ALGO: M (MASTERCARD)
      CHECK IF HOST IS ONLINE ADDR VERIFY: N (Y/N)

***** BASE24 *****
      NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                           F12-HELP
```

CPF Screen 10

CPF screen 10 contains the Stored Value configuration information. This information includes the expiration date for cards activated for the card prefix, and a duration field used to calculate the expiration date for cards activated for the card prefix. The expiration date (specific month and year) and the duration value (number of months to add to current date) are two separate methods of determining the expiration date of the card; either method can be used. Also included is the escheatment field that is used by the escheatment process.

Note: The expiration date from the CAF is passed back to the device handler in the Stored Value token for all Stored Value transactions.

A sample of CPF screen 10 is shown below, followed by descriptions of its fields.

```

BASE24-SV      CARD PREFIX      LLLL  FIID  YY/MM/DD  HH:MM  10 OF 22
PREFIX:                PAN LENGTH: 00                FIID: IDF1

                STORED VALUE INFORMATION

        MIN LOAD LIMIT:                0    MAX LOAD LIMIT:                0

                MAX BALANCE AS CASH:                0

        MAX BALANCE AS CASH REDEMPTION TRAN:                0

                MAX CARD BALANCE:                0

        EXP DATE(Yymm): 0000    OR    DURATION (IN MONTHS):    0

                RETAILER ID:
                ESCHEATMENT: Y
        CAF TEMPLATE LENGTH:  1

***** BASE24 *****
NEW PAGE:                FILE DESTINATION:                NEW LOGICAL NETWORK ID:
                        F12-HELP

```

STORED VALUE INFORMATION

MIN LOAD LIMIT — The minimum amount the account must have when a card activation transaction is performed. This field, along with the MAX LOAD LIMIT field, establishes the range of amounts allowed for an account when it is activated.

Field Length: 1-15 numeric characters depending on the currency used
Required Field: Yes
Default Value: 0
Data Name: CPF.SVCPF.MIN-LOAD

MAX LOAD LIMIT — The maximum amount that the account must have when a card activation transaction is performed. This field, along with the MIN LOAD LIMIT field, establishes the range of amounts allowed for an account when it is activated.

Field Length: 1-15 numeric characters depending on currency used
Required Field: Yes
Default Value: 0
Data Name: CPF.SVCPF-MAX-LOAD

MAX BALANCE AS CASH — The maximum balance remaining that a cardholder can take as a cash advance or purchase with cash back.

Field Length: 1-15 numeric characters depending on the currency used
Required Field: Yes
Default Value: 0
Data Name: CPF.SVCPF.MAX-AS-CASH

MAX BALANCE AS CASH REDEMPTION TRAN — The maximum balance remaining that a cardholder can take as a full redemption.

Field Length: 1-15 numeric characters depending on the currency used
Required Field: Yes
Default Value: 0
Data Name: CPF.SVCPF.MAX-AS-CASH-REDEMPTION

MAX CARD BALANCE — This field represents the maximum balance a cardholder can have in their account.

Field Length: 1-15 numeric characters depending on currency being used
Required Field: Yes
Default Value: 0
Data Name: CPF.POS.SV-MAX-CRD-BAL

EXP DATE — A set expiration date in (MMYY format) placed on the card during a card activation transaction.

Field Length: 4 alphanumeric characters
Required Field: Yes, if the EXP DURATION field is not used.
Default Value: 0000
Data Name: CPF.SVCPF.CRD-EXP-DAT

EXP DURATION — This field is used to calculate the expiration date that can be placed on the card during activation. It represents the number of months to add to the current month.

Field Length: 2 numeric characters
Required Field: Yes, if the EXP DATE field is not used.
Default Value: 0
Data Name: CPF.SVCPF.CRD-DURATION

RETAILER ID — This field is reserved for future use by the Net Position Report.

Field Length: 19 characters
Data Name: CPF.SVCPF.RETAILER-ID

ESCHEATMENT — This field is used by the escheatment process to determine if a card prefix should be escheated by the BASE24 process.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.SV-ESCHEATMENT-FLG

CAF TEMPLATE LENGTH — The number of leading digits of the PAN to use when reading a CAF template record during a card activation transaction.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 1
Data Name: CPF.SVCPF.CAF-TPLT-LGTH

Cardholder Authorization File (CAF)

The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive, Positive Balance, or Parametric Authorization method. The CAF records contain authorization parameters and usage accumulation information for the card issuer's cardholders and are used in authorizing transaction requests.

Note: The Stored Value add-on product supports the Positive Balance method only.

CAF Screen 10

CAF screen 10 contains the BASE24-pos card usage and activity. A sample CAF screen is shown below, with the fields related to configuration of the Stored Value add-on product highlighted. Refer to the ***BASE24 Base Files Maintenance Manual*** for descriptions of these fields.

```

BASE24-pos  CARDHOLDER FILE          LLLL  FIID  YY/MM/DD  HH:MM  10 OF 20

      PAN:                               MEMBER: 000      FIID: BK01

                                POS CARD USAGE CONTROL

                                ACTIVITY LIMITS
                                TOTAL      OFFLINE
CASH WDL:                      0          0
CASH ADV:                      0          0
  AGGR:                        0          0
PURCHASE:                      0          0
RFND/REPL:                  0          0

NUMBER OF RFND/REPL THIS PERIOD:  0      TRAN TC:
      TIMES USED THIS PERIOD:         0      REASON CODE:  (***** )
      TIMES USED PER PERIOD LIMIT:    1      LAST USED DATE:
      ISSUER TXN PROFILE:

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12 - HELP
  
```

Extract Configuration File (ECF)

Extract Configuration File (ECF) records are used to define processing parameters for each type of extract that an institution might perform in a particular logical network. The ECF provides users with the capability to extract a single file or multiple files to a tape or disk file. Users have the option to set parameters that make it possible to perform extracts automatically on a timed basis each day.

The ECF contains a user input flag for the Stored Value History File (SVHF). This flag determines whether a subsequent extract will include the Stored Value History File (SVHF).

ECF Screen 7

ECF screen 7 contains parameters for BASE24-pos extracting processing and PTLF and PRDF extracts. It also contains the parameters for the Stored Value History File (SVHF). SVHF records are extracted to tape if the SVHF record log date and time fall within the ECF starting and ending date and time range. Placing a value of all zeros in the Start Date, Time, and End Date fields on the ECF screen 7 causes an automatic SVHF extract for the previous calendar day. A sample ECF screen 7 is shown below with the SVHF fields highlighted. Refer to the ***BASE24 Base Files Maintenance Manual*** for descriptions of these fields.

```
BASE24-pos  ECF MESSAGE FILE          LLLL          YY/MM/DD  HH:MM  07 OF 23

          TAG:                      SYMBOLIC NAME:

                                POS FILES

                                RELEASE NUM: 60

PTLF: N (Y OR N)  DATE OFFSET:      ( DAYS)
                  GROUP NAME: ALL
REPORT LOCATION:
                  FORMAT: 00 (FIXED FORMAT)
PRDF: N (Y OR N)

SVHF: N (Y OR N)  START DATE:      (YYMMDD) TIME:      (HHMM)
                  END DATE:      (YYMMDD) TIME:      (HHMM)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Extract Screen (EXTR)

The Extract screens provide for direct initiation of file extracts and re-extracts from a series of function key-driven screens. Each screen contains a list of files that can be extracted.

Extract Screen 7

Extract screen 7 is used to control manual extracts of files used solely by the BASE24-pos product. A sample Extract screen 7 is shown below, with the fields related to the Stored Value add-on product highlighted. Refer to the **BASE24 Refresh and Extract Operators Manual** for field descriptions.

```
BASE24-pos    EXTRACT SCREENS          LLLL          YY/MM/DD  HH:MM  07 OF 23

SYMBOLIC NAME:                ( OR RN NAME AND PROCESS NUMBER )
      RN NAME:                PROCESS NUMBER:
      ECF KEY:                OVERRIDE ECF: N

                                POS FILES

                                PTLF:  N
                                PRDF:  N
                                SVHF:  N

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F1-VALIDATE      F2-EXTRACT    F3-REEXTRACT    F12-HELP
```

Institution Definition File (IDF)

The Institution Definition File (IDF) contains one record for each institution participating in the logical network and defines processing for each institution. The IDF contains routing tables for transaction routing within a BASE24 product and the parameters for cards, dates, processing control and sharing.

IDF Screen 1

IDF screen 1 contains fields identifying the financial institution and file names. A sample IDF screen 1 is shown below, with fields relevant to configuration of the Stored Value add-on product highlighted. Refer to the *BASE24 Base Files Maintenance Manual* for descriptions of its fields.

```
BASE24-BASE  INSTITUTION FILE          LLLL          YY/MM/DD  HH:MM  01 OF 42

          FIID:          FI-NAME:

STATE:    0  COUNTY:    0  COUNTRY:  840  PHONE:
INSTITUTION ID NUMBER:  00000000000  REFRESH GROUP:

          FILE NAMES:

          NEG:
          UAF:
          CAF:
          SPF:
          PBF1:
          PBF2:
          PBF3:
PBF4:

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

                                0000
```

POS Transaction Log File (PTLF)

The PTLF provides an audit trail for activity occurring at POS terminals. All financial transactions occurring for each processing day at POS terminals throughout the processing day at POS terminals throughout the logical network are logged to the PTLF. This enables retailers and processors to track cardholder, retailer, or terminal activity.

This file can be perused online for transactions initiated by a specific retailer, terminal or operator.

PTLF Screen 4

PTLF Perusal Detail Screen 4 displays the Track 2 data from the Stored Value token when the data is present. The token data is read when the transaction is an Additional Card Activation transaction. A sample PTLF screen 4 is shown below with fields specific to the Stored Value add-on product highlighted. Refer to the *BASE24-pos Files Maintenance Manual* for descriptions of its fields.

```

BASE24-pos    PTLF DETAIL                LLLL                YY/MM/DD  HH:MM  04 OF 04

PERUSAL TYPE/REC FRMT:  D  DATE(YYMMDD):  980529  FIID:  FIID  LLLL:  LLLL
CARD:  1234567890123456789                MEMBER NUM:  000
TERMINAL:  CNTRLT00                RETAILER:  RETAILER01                CLERK:  MANDY

      PAN: 1234567890123456789    MEMBER NUM: 000    EXP DATE: 0000
CARD TYPE: P                ISS CODE: 00    BATCH NUM: 000
TERM: CNTRLT00                SIC CODE: 0040    SHIFT NUM: 000
TRAN DATE: 05/29                TIME OFST: 0    TRAN TIME: 10:08:22
TRAN: 262010 ADNL CRD ACT    ERR FLAG:    PIN PRESENT: N
DRAFT CAP: 0 (AUTH ONLY    )    APPRV CODE: 286743 B    INVOICE: 0000000000
ACCT: 1234567890123456789    OVERRIDE FLAG: 0    ORIG INVOICE: 0000000000
AUTHORIZER: AUTHH                AUTH IND: P    SEQ NUM: 000000000023
RESP CODE: 5001 (APPROVED-NO BALANCES)    AUTH SEQ NUM: 000000000000
      AMT 1:                0.00                INDUSTRY FRMT:
      AMT 2:                US
ALT MRCHNT:
ADDITIONAL CARD NUMBER: 1123456789012345678

***** BASE24 *****
      FILE DESTINATION:                NEW LOGICAL NETWORK ID:
F5-GOTO SUMMARY  F6-NXT REC  F7-PRV REC  F9-NXT PAGE  F11-PRV PAGE  F12-HELP

```

ADDITIONAL CARD NUMBER — The card number of a stored value card on which an additional card activation transaction was performed. This field is displayed only when the transaction is a BASE24-pos Stored Value additional card activation transaction. It is not displayed for other BASE24-pos Stored Value transactions, standard transactions or transactions using secure Internet validation.

Field Length: 19 alphanumeric characters

Data Name: Not applicable

Positive Balance File (PBF)

The Positive Balance File (PBF) contains one record for each account belonging to a BASE24-pos cardholder whose card issuer uses the Positive Balance or Parametric Authorization method.

The PBF displays various balances and amounts used by BASE24-pos to authorize transaction requests without having to forward them to a back-end host authorization system.

PBF screen 1 contains the following three fields to support the BASE24-pos Stored Value product dormancy processing:

- Balance Prior to Dormancy Date. This is the PBF balance before the dormancy fee is assessed.
- Dormancy Date. This is the date a stored value dormancy fee is assessed.
- Card Activation State. This is the stored value card activation state.

Refer to the ***BASE24 Base Files Maintenance Manual*** for a description of the PBF and these stored value fields.

Stored Value History File (SVHF)

The Stored Value History File (SVHF) is an Enscribe-based, key-sequenced file that contains a summary of all financial transactions involving Stored Value cards. Access to this file is done using the SVHF requester and server pair.

If transactions are routed to an interchange for authorization, logging to this file is optional. The Router Authorization Process (RTAU) retrieves the SVHF-ICHG-AUTH-LOG param from the Logical Network Configuration File (LCONF) upon initialization or warmboot.

The SVHF is created and written to, whenever a PTLF record is added for an approved, denied, or reversed transaction (other than a balance inquiry). Records are retained in this file until they are deleted by the History File Cleanup module (HFCLUP) based on the value of the POS-SVHF-RET-PERIOD LCONF param. The SVHF records are keyed to the account number to enable operators to have read access to all applicable history data on the common account. Both Retailer ID and Terminal ID keys identify the keys to the record.

You can peruse this file using any of the following methods:

- Card number
- Card number with a beginning and ending date
- Card number and transaction type
- Card number and transaction type with a beginning and ending date
- Retailer ID
- Retailer ID with a beginning and ending date
- Retailer and terminal ID
- Retailer and terminal ID with a beginning and ending date

All of the detail information is presented in reverse chronological order (i.e., the most recent transaction is presented first).

You can access the SVHF by selecting the SVHF entry from the POS Product Menu or by entering SVHF in the FILE DESTINATION field of a BASE24 screen and pressing the **F16** key. This screen allows eight types of perusal of the Stored Value History File.

The first perusal type is card perusal by card number only. This requires that no beginning or ending dates be entered. The FIID and CARD fields must be filled.

The second perusal type is card perusal by card number using a beginning and an ending date. In this case, the beginning and ending dates must be filled along with the FIID and CARD fields.

The third perusal type is transaction code perusal only. This allows perusal by transaction code as a primary and card number as secondary. The FIID, TRAN CODE, and CARD fields must be filled.

The fourth perusal type is transaction code perusal using beginning and ending dates. In this case, the beginning and ending fields must be filled along with the FIID, TRAN CODE, and CARD fields.

The fifth perusal is retailer perusal only. This type allows perusal by retailer only with no beginning and ending dates. The RETAILER ID must be filled.

The sixth perusal type is retailer using a retailer ID and beginning and ending dates. In this case, the beginning and ending fields must be filled along with the RETAILER ID field.

The seventh perusal type is retailer and terminal perusal by retailer and terminal only. This type allows perusal by retailer and terminal with no beginning and ending dates.

The eighth perusal is perusal using retailer ID, terminal ID and beginning and ending dates. In this case, the beginning and ending fields must be filled along with the RETAILER and TERMINAL fields.

The following pages describe how these screens are accessed. They also show an example of each screen format and describe the information on each screen.

Key Screen Function Keys

Some function keys for these screens differ from the standard function keys explained in the *BASE24 Base Files Maintenance Manual* and the *BASE24 CRT Access Manual*. The use of these function keys is explained in the table below.

The first column shows the BASE24 function keys. The second column provides descriptions of how the function keys work on the SVHF key screens.

Key	Description
F1	Card Number — Accesses the summary screen for card number perusal. The summary screen is displayed but only the key fields contain data. Before using this function key, you must first complete the fields in the CARD NUMBER PERUSAL portion of the SVHF key screen. This function key is valid only when used from the SVHF key screen 1.
F2	Tran Code — Accesses the summary screen for tran code perusal. The summary screen is displayed but only the key fields contain data. Before using this function key, you must first complete the fields in the TRAN CODE PERUSAL portion of the SVHF key screen. This function key is valid only when used from SVHF key screen 1.
F3	Retailer — Accesses the summary screen for retailer perusal. The summary screen is displayed but only the key fields contain data. Before using this function key, you must first complete the fields in the RETAILER PERUSAL portion of the SVHF key screen. This function key is valid only when used from the SVHF key screen 1.
F4	Retailer or Terminal — Accesses the summary screen for retailer or terminal perusal. The summary screen is displayed but only the key fields contain data. Before using this function key, you must first complete the fields in the RETAILER or TERMINAL PERUSAL portion of the SVHF key screen. This function key is valid only when used from the SVHF screen 1.

SVHF Key Perusal Selection Screen 1

SVHF key screen 1 can be displayed if the system is configured for SVHF perusal and the operator has access to SVHF key screen 1 in the Security File (SEC). The type of perusal performed (card number, tran code, retailer, or retailer and terminal) depends on the fields the operator enters on the key screen.

To access SVHF key screen 1, place the cursor before SVHF on the POS Product Menu screen and press the **F1** function key.

The SVHF key screen 1 is shown below, followed by descriptions of its fields.

```

BASE24-pos   SVHF KEY                               LLLL       YY/MM/DD  HH:MM  01 OF 03

*****
CARD NUMBER  TYP: C  FIID: _____  BEGIN DATE (YYYYMMDD) : _____
PERUSAL                                           END DATE (YYYYMMDD) : _____
                                CARD: _____

*****
TRAN CODE    TYP: T  FIID: _____  BEGIN DATE (YYYYMMDD) : _____
PERUSAL      TRAN CODE: _____  END DATE (YYYYMMDD) : _____
                                CARD: _____

*****
RETAILER     TYP: R  FIID: _____  BEGIN DATE (YYYYMMDD) : _____
PERUSAL                                           END DATE (YYYYMMDD) : _____
                                RETAILER: _____

*****
RETL/TERM    TYP: D  FIID: _____  BEGIN DATE (YYYYMMDD) : _____
PERUSAL      RETAILER: _____  END DATE (YYYYMMDD) : _____
                                TERMINAL: _____

*****
***** BASE24 *****
                                FILE DESTINATION:  NEW LOGICAL NETWORK ID:
F1-CARD      F2-TRAN CODE    F3-RETAILER    F4-RETAILER/TERMINAL    F12-HELP
  
```

CARD NUMBER PERUSAL

The following fields are used to select specific transactions for card perusal. Records are displayed based on the FIID of the card-issuing institution, the dates of the transactions, and the cardholder number.

TYP — A code indicating the type of transaction being perused. Valid values are as follows:

C = Card
T = Tran Code
R = Retailer
D = Retailer and Terminal

Field Length: 1 alphabetic character
Required Field: No
Default Value: C
Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: 1-4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.PRIKEY.FIID

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Current date
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Current date
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

CARD — The number of the card that initiated the Stored Value transactions being selected for perusal.

Field Length: 19 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.PRIKEY.CARD-NUM

TRAN CODE PERUSAL

The following fields are used to select specific transactions for transaction code perusal. Records are displayed based on the FIID of the card-issuing institution, the transaction code, the cardholder number, and the date of the transaction.

TYP — A code indicating the type of transaction being perused. Valid values are as follows:

C = Card
T = Tran Code
R = Retailer
D = Retailer and Terminal

Field Length: 1 alphabetic character
Required Field: No
Default Value: T
Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: 1-4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.ALTKEY1.FIID

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Blanks
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

TRAN CODE — A code identifying the type of transaction. Valid values are as follows:

10 = Normal purchase
11 = Preauthorization purchase
12 = Preauthorization purchase completion
13 = Mail or phone order
14 = Merchandise return
15 = Cash advance
16 = Card verification
17 = Balance inquiry (not logged)
18 = Purchase with cash back
19 = Check verification
20 = Check guarantee
21 = Purchase adjustment
22 = Adjustment - merchandise return
23 = Adjustment - cash advance
24 = Adjustment - purchase with cash back
25 = Stored value card activation
26 = Stored value additional card activation
27 = Stored value replenishment transaction
28 = Stored value full redemption

Field Length: 2 numeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.ALTKEY1.TRAN-CDE

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Current date
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

CARD — The number of the card that initiated the Stored Value transactions being selected for perusal.

Field Length: 19 alphanumeric characters
Required Field: Yes
Default Value: No default values
Data Name: SVHF.ALTKEY1.CRD-NUM

RETAILER PERUSAL

The following fields are used to select specific transactions for retailer perusal. Records are displayed based on the FIID of the card-issuing institution, transaction date and the retailer ID.

TYP — A code indicating the type of transaction being perused. Valid values are as follows:

C = Card
T = Tran Code
R = Retailer
D = Retailer and Terminal

Field Length: 1 alphabetic character
Required Field: No
Default Value: R
Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: 1-4 alphanumeric characters
Required Field: Yes
Default Value: No default values
Data Name: SVHF.ALTKEY2.FIID

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8
Required Field: No
Default Value: Blanks
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8
Required Field: No
Default Value: Blanks
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

RETAILER — The retailer ID being used to peruse the Stored Value transactions.

Field Length: 1-19 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.ALTKEY2.RETL-ID

RETAILER or TERMINAL PERUSAL

The following fields are used to select specific transactions for retailer and terminal perusal. Records are displayed based on the FIID of the card-issuing institution, the retailer ID, the terminal ID, and the date of the transaction.

TYP — A code indicating the type of transaction being perused. Valid values are as follows:

C = Card
T = Tran Code
R = Retailer
D = Retailer and Terminal

Field Length: 1 alphabetic character
Required Field: None
Default Value: D
Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: 1-4 alphanumeric characters
Required Field: None
Default Value: No default value
Data Name: SVHF.ALTKEY3.FIID

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Current date
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

RETAILER — The retailer ID being used to peruse the Stored Value transactions.

Field Length: 1-19 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.ALTKEY3.RETL-ID

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: 8 numeric characters
Required Field: No
Default Value: Current date
Data Name: SVHF.PRIKEY.TRAN-DAT-TIM

TERMINAL — The terminal ID of the terminal owner used to access the appropriate SVHF record.

Field Length: 1-16 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: SVHF.ALTKEY3.TERM-ID

BEGIN DATE — The beginning date of the SVHF for which transactions are displayed.

Field Length: System protected
Data Name: Not applicable

CARD — The card number of the card that initiated the transactions selected for perusal.

Field Length: System protected
Data Name: SVHF.PRIKEY.CARD-NUM

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

MBR NUM — The member number or card sequence number for the card that initiated the transaction. When multiple cards are issued with the same card number, the value in this field distinguishes between the cards. If an institution does not use member numbers, the value in this field is 000.

Field Length: System protected
Data Name: SVHF.DETL.MBR-NUM

REF NUM — The reference number is a unique identifier for each transaction in the file. Stored value transactions are identified with a reference number beginning with “B24”.

Field Length: System protected
Data Name: SVHF.DETL.REF-NUM

TRAN CODE — A code that identifies the type of transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY1.TRAN-CDE

RESP CODE — The response code for the transaction.

Field Length: System protected
Data Name: SVHF.DETL.RESP-CDE

TRAN DATE — The date the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN TIME — The time the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN AMOUNT — The amount of the transaction in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.ALTKEY3.AMT

D/C IND — The Debit or Credit indicator for the amount of the transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY3.DB-CR-IND

ACCOUNT BALANCE — The balance of the Stored Value card after the current transaction, in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.ALTKEY3.BAL

SVHF Detail Screen 3 — Format 1

The SVHF Detail screen 3—format 1 is displayed from the SVHF Card Summary screen when and the perusal type is C (card). SVHF Detail screen 3 is shown below, followed by descriptions of its corresponding fields.

```

BASE24-pos   SVHF DETAIL                LLLL      YY/MM/DD  HH:MM  03 OF 03

  TYP:  _ (C, T, R or D)

RETAILER ID:  _____
TERMINAL ID   :  _____

CARD NUMBER   :  _____
MEMBER NUMBER :  _____
TRAN DATE     :  _____
TRAN TIME     :  _____
TRAN AMOUNT   :  _____ -
DEBIT/CREDIT IND:  _ (DESC  )
TRAN CODE     :  _ (TRAN DESCRIPTION )
RESPONSE CODE :  _ (RESPONSE DESCRIPTION )
REFERENCE NUM :  _____
ACCOUNT BALANCE :  _____

***** BASE24 *****
FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F5-GOTO SUMMARY           F8-GOTO KEY SCREEN           F12-HELP
                                                                0000

```

TYP — A code indicating the type of transaction being perused.

Field Length: System protected

Data Name: Not applicable

RETAILER ID — The retailer ID used to peruse the Stored Value transactions.

Field Length: System protected

Data Name: SVHF.ALTKEY2.RETL-ID

TERMINAL ID — The terminal ID of the terminal owner used to access the appropriate SVHF record.

Field Length: System protected

Data Name: SVHF.ALTKEY3.TERM-ID

CARD NUMBER — The card number of the card that initiated the transactions selected for perusal.

Field Length: System protected
Data Name: SVHF.ALTKEY1.CARD-NUM

MEMBER NUMBER — The member number or card sequence number for the card that initiated the transaction. When multiple cards are issued with the same card number, the value in this field distinguishes between the cards. If an institution does not use member numbers, the value in this field is 000.

Field Length: System protected
Data Name: SVHF.DETL.MBR-NUM

TRAN DATE — The date the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN TIME — The time the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN AMOUNT — The amount of the transaction in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.DETL.AMT

DEBIT/CREDIT IND — The debit or credit indicator for the amount of the transaction.

Field Length: System protected
Data Name: SVHF.DETL.DB-CR-IND

TRAN CODE — A code that identifies the type of transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY1.TRAN-CDE

RESPONSE CODE — The response code for the transaction.

Field Length: System protected
Data Name: SVHF.DETL.RESP-CDE

REFERENCE NUM — The reference number is a unique identifier for each transaction in the file. Stored value transactions are identified with a reference number beginning with “B24”.

Field Length: System protected
Data Name: SVHF.DETL.REF-NUM

ACCOUNT BALANCE — The balance of the Stored Value card after the current transaction, in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.DETL.BAL

SVHF Transaction Code Summary List Screen 2

SVHF Transaction Code Summary list screen 2 can be displayed if the operator has security access to SVHF screen 2. The SVHF Transaction Code Summary List screen is accessed by pressing the **F1** key from the SVHF key Perusal Selection screen. SVHF Transaction Code Summary screen 2 is shown below, followed by descriptions of its corresponding fields.

Note: After you access the SVHF Transaction Code Summary List screen 2, you must press the **F1** key to display the first page of records summary lines.

```

BASE24-pos   SVHF TC SUMMARY LIST      LLLL      YY/MM/DD  HH:MM  02 OF 03

TYP: T  FIID:  ____  TRAN CODE:  ____ (      )  BEGIN DATE :  ____
        CARD:  ____                        END DATE :  ____

      MBR      REF      RESP      TRAN      TRAN      TRAN      D/C      ACCOUNT
      NUM      NUMBER   CODE      DATE      TIME      AMOUNT   IND      BALANCE
      - - - - -

```



```

***** BASE24 *****
FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F1-FIRST PAGE      F4-DETAIL      F9-NEXT PAGE      F11-PREV PAGE      F12-HELP 0000

```

TYP — A code indicating the type of transaction being perused. Valid value is T (transaction code perusal).

Field Length: System protected

Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: System protected

Data Name: SVHF.PRIKEY.FIID

TRAN CODE — A code that identifies the type of transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY1.TRAN-CDE

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

CARD — The card number of the card that initiated the transactions selected for perusal.

Field Length: System protected
Data Name: SVHF.PRIKEY.CARD-NUM

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

MBR NUM — The member number or card sequence number for the card that initiated the transaction. When multiple cards are issued with the same card number, the value in this field distinguishes between the cards. If an institution does not use member numbers, the value in this field is 000.

Field Length: System protected
Data Name: SVHF.DETL.MBR-NUM

REF NUM — The reference number is a unique identifier for each transaction in the file. Stored value transactions are identified with a reference number beginning with “B24”.

Field Length: System protected
Data Name: SVHF.DETL.REF-NUM

RESP CODE — The response code for the transaction.

Field Length: System protected
Data Name: SVHF.DETL.RESP-CDE

TRAN DATE — The date the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN TIME — The time the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN AMOUNT — The amount of the transaction in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.ALTKEY3.AMT

D/C IND — The debit or credit indicator for the amount of the transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY3.DB-CR-IND

ACCOUNT BALANCE — The balance of the Stored Value card after the current transaction, in whole and fractional currency units.

Field Length: System protected
Data Name: SVHF.ALTKEY3.BAL

SVHF Detail Screen 3—Format 2

SVHF Detail screen 3—format 2 is displayed from the SVHF Transaction Code Summary screen when the perusal type is T (transaction code). The SVHF Detail screen 3—format 2 is shown below. For the corresponding field information, refer to the SVHF Detail screen 3—format 1 explained earlier in this section.

```
BASE24-pos   SVHF DETAIL           LLLL           YY/MM/DD  HH:MM  03 OF 03

  TYP:  _ (C, T, R or D)

RETAILER ID:  _____
TERMINAL ID   :  _____

CARD NUMBER   :  _____
MEMBER NUMBER :  _____
TRAN DATE     :  _____
TRAN TIME     :  _____
TRAN AMOUNT   :  _____ -
DEBIT/CREDIT IND:  _ (DESC  )
TRAN CODE     :  _ (TRAN DESCRIPTION )
RESPONSE CODE :  _ (RESPONSE DESCRIPTION )
REFERENCE NUM :  _____
ACCOUNT BALANCE :  _____

***** BASE24 *****
FILE DESTINATION: NEW LOGICAL NETWORK ID:
F5-GOTO SUMMARY  F8-GOTO KEY SCREEN      F12-HELP
                                           0000
```

SVHF Retailer Summary List Screen 2

SVHF Retailer Summary List screen 2 can be displayed if the operator has security access to SVHF screen 2. The SVHF Retailer Summary List screen is accessed by pressing the **F1** key from the SVHF Key Perusal Selection screen. SVHF Retailer Summary screen 2 is shown below, followed by descriptions of its corresponding fields.

Note: After you access the SVHF Retailer Summary List screen, you must press the **F1** key to display the first page of records summary lines.

```

BASE24-pos   SVHF RETL SUMMARY LIST   LLLL           YY/MM/DD  HH:MM  02 OF 03

TYP: R
RETAILER ID: _____

BEGIN DATE: _____
END DATE:   _____

CARD NUMBER      MBR  TRAN  RESP      TRAN      TRAN
                  NUM  CODE  CODE      DATE      TIME
-  _____  -  -  -  -  -  -  -

***** BASE24 *****
FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F1-FIRST PAGE  F2-LAST PAGE  F4-DETAIL  F9-NEXT PAGE  F11-PREV PAGE  F12-HELP
                                           0000
  
```

TYP — A code indicating the type of transaction being perused. Valid value is T (transaction code perusal).

Field Length: System protected

Data Name: Not applicable

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected

Data Name: Not applicable

RETAILER ID — The retailer ID used to peruse the Stored Value transactions.

Field Length: System protected
Data Name: SVHF.ALTKEY2.RETL-ID

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

CARD NUMBER — The card number of the card that initiated the transactions selected for perusal.

Field Length: System protected
Data Name: SVHF.ALTKEY1.CARD-NUM

MBR NUM — The member number or card sequence number for the card that initiated the transaction. When multiple cards are issued with the same card number, the value in this field distinguishes between the cards. If an institution does not use member numbers, the value in this field is 000.

Field Length: System protected
Data Name: SVHF.DETL.MBR-NUM

TRAN CODE — A code that identifies the type of transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY1.TRAN-CDE

RESP CODE — The response code for the transaction.

Field Length: System protected
Data Name: SVHF.DETL.RESP-CDE

TRAN DATE — The date the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN TIME — The time the record was logged.

Field Length: System protected

Data Name: Not applicable

SVHF Detail Screen 3—Format 3

The SVHF Detail Screen 3—format 3 is displayed from the SVHF Retailer Summary screen when the perusal type is R (retailer).

The SVHF Detail screen is shown below. For the corresponding field information, refer to SVHF Detail screen 3—format 1 explained earlier in this section.

```
BASE24-pos    SVHF DETAIL                LLLL      YY/MM/DD  HH:MM  03 OF 03

  TYP:  _ (C, T, R or D)

RETAILER ID:  _____
TERMINAL ID   :  _____

CARD NUMBER   :  _____
MEMBER NUMBER :  _____
TRAN DATE     :  _____
TRAN TIME     :  _____
TRAN AMOUNT   :  _____ -
DEBIT/CREDIT IND:  _ (DESC  )
TRAN CODE     :  _ (TRAN DESCRIPTION )
RESPONSE CODE :  _ (RESPONSE DESCRIPTION )
REFERENCE NUM :  _____
ACCOUNT BALANCE :  _____

***** BASE24 *****
FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F5-GOTO SUMMARY           F8-GOTO KEY SCREEN           F12-HELP
                                0000
```

SVHF Retailer and Terminal Summary List Screen 2

SVHF Retailer and Terminal Summary List screen 2 can be displayed if the operator has security access to SVHF screen 2. The SVHF Retailer and Summary screen is accessed by pressing the **F1** key from the SVHF Key Perusal Selection screen. SVHF TERM Summary screen 2 is shown below followed by descriptions of its corresponding fields.

Note: After you access the SVHF Retailer and Summary screen 2, you must press the **F1** key to display the first page of records summary lines.

```

BASE24-pos   SVHF TERM SUMMARY LIST   LLLL           YY/MM/DD  HH:MM  02 OF 03

TYP: D  FIID: _____  RETAILER ID: _____  BEGIN DATE: _____
                        TERMINAL ID: _____  END DATE: _____

      CARD NUMBER           MBR  TRAN  RESP      TRAN      TRAN
                           NUM  CODE  CODE      DATE      TIME
      - _____  -    -    -    -    -    -

*****
                        FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F1-FIRST PAGE      F4-DETAIL      F9-NEXT PAGE      F11-PREV PAGE      F12-HELP
                                                                0000
  
```

TYP — A code indicating the type of transaction being perused. Valid value is D (retailer and terminal perusal).

Field Length: System protected
Data Name: Not applicable

FIID — The financial institution identifier of the card-issuing institution. The value in this field is used to select the correct SVHF records.

Field Length: System protected
Data Name: SVHF.PRIKEY.FIID

RETAILER ID — The retailer ID being used to peruse the Stored Value transactions.

Field Length: 19
Required Field: Yes
Default Value: Blanks
Data Name: SVHF.ALTKEY3.RETL-ID

BEGIN DATE — The beginning date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

TERMINAL ID — The terminal ID of the terminal owner used to access the appropriate SVHF record.

Field Length: System protected
Data Name: SVHF.ALTKEY3.TERM-ID

END DATE — The ending date (YYYYMMDD) of the date range of SVHF records to be displayed.

Field Length: System protected
Data Name: Not applicable

CARD NUMBER — The card number of the card that initiated the transactions being selected for perusal.

Field Length: System protected
Data Name: SVHF.PRIKEY.CARD-NUM

MBR NUM — The member number or card sequence number for the card that initiated the transaction. When multiple cards are issued with the same card number, the value in this field distinguishes between the cards. If an institution is not using member numbers, the value in this field will be 000.

Field Length: System protected
Data Name: SVHF.DETL.MBR-NUM

TRAN CODE — A code that identifies the type of transaction.

Field Length: System protected
Data Name: SVHF.ALTKEY1.TRAN-CDE

RESP CODE — The response code for the transaction.

Field Length: System protected
Data Name: SVHF.DETL.RESP-CDE

TRAN DATE — The date the record was logged.

Field Length: System protected
Data Name: Not applicable

TRAN TIME — The time the record was logged.

Field Length: System protected
Data Name: Not applicable

SVHF Detail Screen 3—Format 4

The SVHF Detail screen 3—format 4 is displayed from the SVHF Retailer and Terminal Summary screen when the perusal type is D (retailer and terminal).

The SVHF Detail screen 3—format 4 is shown below. For the corresponding field information, refer to the SVHF Detail screen 3—format 1 explained earlier in this section.

```
BASE24-pos   SVHF DETAIL                LLLL      YY/MM/DD  HH:MM  03 OF 03

  TYP: _ (C, T, R or D)

RETAILER ID: _____
TERMINAL ID   : _____

CARD NUMBER   : _____
MEMBER NUMBER : _____
TRAN DATE     : _____
TRAN TIME     : _____
TRAN AMOUNT   : _____ -
DEBIT/CREDIT IND: _ (DESC )
TRAN CODE     : _ (TRAN DESCRIPTION )
RESPONSE CODE : _ (RESPONSE DESCRIPTION )
REFERENCE NUM : _____
ACCOUNT BALANCE : _____

***** BASE24 *****
FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F5-GOTO SUMMARY           F8-GOTO KEY SCREEN           F12-HELP
                                0000
```

Stored Value Dormancy File (SVDF)

The Stored Value Dormancy File (SVDF) is an Enscribe-based, key-sequenced file that contains the fees that can be accessed for an account that has been determined to be dormant.

BASE24 customers that support stored value cards have the ability to apply dormancy fees to cards that have not expired but have not been used for a period of time. The card issuer can collect fees, take the account balance to zero, and eventually clear the stored value card from the system.

The Dormancy/Escheatment process reads the SVDF, applies any applicable fees to the corresponding Positive Balance File (PBF) accounts, logs the fees to the Stored Value History File (SVHF), and creates a summary report for the fees applied.

You can access the SVDF by selecting the SVDF entry from the POS Product Menu or by entering SVDF in the FILE DESTINATION field of a BASE24 screen and pressing the **F16** key.

The SVDF screen is shown below, followed by descriptions of its fields.

```

BASE24-POS   DORMANCY               LLLL      YY/MM/DD  HH:MM  01 OF 01
RETAILER ID: *****              CARD TYPE: ** (ALL      )
ACTIVATION STATE: ***

          FLAT FEE:      0.00          OR PERCENT FEE:  0.00

DAYS DORMANT:  365
PERIOD:       M (MONTHLY      )

***** BASE24 *****
          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
          F12 - HELP

```

RETAILER ID — The retailer identifier. The value in this field must match the retailer ID in the Card Prefix File (CPF).

Field Length: 19 alphanumeric characters
Required Field: Yes
Default Value: Asterisks
Data Name: CPF.SVDF.PRIKEY.RETAILER-ID

CARD TYPE — The stored value card type.

Field Length: 2 alphanumeric characters
Required Field: Yes
Default Value: Asterisks
Data Name: SVDF.PRIKEY.CRD-TYP

ACTIVATION STATE — The state where the stored value card is activated.

Field Length: 3 alphanumeric characters
Required Field: Yes
Default Value: Asterisks
Data Name: SVDF.PRIKEY.ACT-ST

FLAT FEE — The flat fee that is charged as the dormancy fee. Either a flat fee or percent fee is used as a dormancy fee. If a percent fee is used, this field must be zero.

Field Length: 1–7 numeric characters and one decimal point
Required Field: Yes, if the percent fee is blank.
Default Value: 0
Data Name: SVDF.FLAT-FEE

PERCENT FEE — The percentage of the balance that is charged as the dormancy fee. Either a percent fee or a flat fee is used as a dormancy fee. If a flat fee is used, this field must be zero.

Field Length: 1–4 numeric characters and one decimal point
Required Field: Yes, if the flat fee is blank.
Default Value: 0
Data Name: SVDF.FLAT-FEE

DAYS DORMANT — The number of days during which an account has no activity before a dormancy fee is assessed.

Field Length: 5 numeric characters
Required Field: Yes
Default Value: 365
Data Name: SVDF.DAYS-DORMANT

PERIOD — This field identifies the last file maintenance action on this record. The value in this field includes the user who did the update, time at which it was done, and the type of update. Valid values are as follows:

W = Weekly
BW = Biweekly
A = Annually
Q = Quarterly
S = Semiannually

Field Length: 5 alphanumeric characters
Required Field: Yes
Default Value: M
Data Name: SVDF.PRD

ACI Worldwide Inc.

3: Authorization Processing

This section describes additional authorization processing that is performed on Stored Value transactions. The following topics are described:

- CAF and PBF usages
- Adding CAF records
- Card Activation transaction
- Full Redemption transaction
- Additional Card Activation transaction
- Replenishment transaction
- Offline authorization of stored value cards

Note: Stored Value transactions are authorized using the offline authorization level and the Positive Balance authorization method.

CAF and PBF Usages

The following table describes how certain CAF and PBF usage fields are updated for Stored Value transactions. “A1” refers to the transaction amount found in the AMT-1 fields from the PSTM. A plus sign (+) indicates that the funds are to be added to the usage accumulator. A minus sign (-) indicates that the funds are to be subtracted from the usage accumulator. A “+1” indicated that the accumulator is incremented by 1. All other transactions follow standard processing.

Transaction Description	PBF Totals			CAF Totals			
	Cash Out	Ledger Balance	Available Balance	Times Used	Number of Refunds/ Replenishments	Refund/Replenishments	Cash Withdrawal
Card Activation *		+A1	+A1		+1	+A1	
Card Replenish		+A1	+A1	+1	+1	+A1	
Full Redemption	+A1	-A1	-A1	+1			+A1

* The PBF totals will not be modified for preloaded cards.

Adding CAF Records

The Router/Authorization module reads the CAF template record if a CAF record does not exist for card activation and additional card activation transactions.

Note: The CAF for the existing card must already exist and will not be added. This is applicable for additional card activation transactions only.

The CAF TEMPLATE LENGTH field read from the CPF indicates how many bytes of the PAN are used to read the CAF template record. The primary key of the template CAF record is constructed from the first several bytes of the card number being activated. This consists of the following:

- PAN taken from the Stored Value token (Only on additional card activation transactions. Otherwise, the PAN is in the PSTM.)
- Member Number “000”
- FIID associated with the institution

The following CAF fields in the template CAF record are overwritten:

CAF Field	Data Value
PAN	PSTM.TRAN.TRACK 2[1] for card activation transactions only.
Member Number	TRK2.MBR-NUM for card activation From the field of the Stored Value token for additional card activation transactions.
Card Status	0 (issued not active)
PIN Offset	zeros
First Used Date	Current system date (YYMMDD)
Last Reset Date	Current system date (YYMMDD)
Expiration Date	CPF expiration date value, or calculated from the CPF duration value. Use duration value if non-zero. If duration is present then expiration date is calculated as the current date + duration. (YYMM)
ACCT-Account Count	1
ACCT-Account Type	01 (checking)
ACCT-Account Number	PAN (PSTM.TRAN.TRACK 2[1]) for card activation transactions only. The account number from the existing CAF for additional card activation transactions.
ACCT-Account Status	P (open primary)
ACCT-Account Description	“Stored Val”
ACCT-CORP, ACH-IND	Blanks
POS-Last Used	Current system date (YYMMDD)

All of the other BASE, POS, or other segment fields are defined in the template CAF record.

The complete CAF record for the card being activated is then added into the CAF data file. If the record already exists or cannot be added, it is denied with response code 104.

Card Activation Transaction

When a Card Activation transaction is processed, the Router/Authorization module performs the following steps:

Steps	Processing
1	Verifies that the CAF Card Type value is valid for Stored Value transactions. Valid values are: SR = Stored Value reloadable SN = Stored Value non-reloadable If the Card Type is not valid, the transaction is denied with response code 055.
2	Verifies the amount requested. The amount must be in the range defined by the CPF minimum load limit and the maximum load limit. If the amount is over the maximum load limit, the transaction is denied with response code 095. If the amount is less than the minimum load limit, the transaction is denied with response code 092.
3	Verifies that the number of refunds limit has not been exceeded. If it is exceeded, then the transaction is denied with response code 087.
4	Verifies that the per refund limit will not be exceeded. If exceeded, then the transaction is denied with response code 081.
5	Verifies that the offline refund limit will not be exceeded. If exceeded, then the transaction is denied with response code 080.
6	Verifies that the total refund limit will not be exceeded. If exceeded, then the transaction is denied with response code 083.
7	If the CAF record already exists and the card status is not 0 (Issued not active), the transaction is denied with response code 058.
8	Retrieves the account index. If no index is found, the transaction is denied with response code 060.

Steps**Processing**

- 9 Reads the PBF record associated with the CAF record. If the PBF record does not exist, then a new PBF record is added. If any other problem is detected while reading the PBF, the transaction is denied with response code 060.
- 10 If the PBF record does not exist, then the transaction amount must not be zero and the transaction is declined with response code 069.
- 11 Formats the PBF record as follows:

PBF Field	Setting
PRIKEY.FIID	FIID (from CAF, CPF, or IDF)
PRIKEY.NUM	PAN (PSTM.TRAN.TRACK 2[1])
PRIKEY.TYP	1 (checking)
ACCT-STAT	9 (closed)
AVAIL-BAL	PSTM.TRAN.AMT-1
LEDG-BAL	PSTM.TRAN.AMT-1
AMT-ON-HLD	0
OVRDRFT-LMT	0
LAST-DEP-DAT	Current system date (YYMMDD)
LAST-DEP-AMT	PSTM.TRAN.AMT-1
LAST-WDL-DAT	0
LAST-WDL-AMT	0
CASH-OUT-TODAY	0
CASH-IN-TODAY	0
POS Fields	zeros

- 12 If the PBF record cannot be added or if it already exists, the transaction is denied with response code 062. If a PBF record does exist and the activation amount on the card matches the PBF available balance, the transaction is approved. If a PBF record exists and the activation amount does not match the PBF, the transaction is denied with response code 069. If a PBF record does exist and the activation is zero, the transaction is approved.
- 13 If the PBF record could not be added successfully, clean up should be performed so there isn't a CAF record that is not linked to a PBF record. The associated CAF record needs to be updated to change the card status to 0 (Issued Not Active).

Steps	Processing
14	Verifies that the new balance is not greater than the maximum card balance from the CPF. If the amount is over the limit, the transaction is denied with response code 095 (Amount over maximum).
15	Updates the PBF usage accumulator fields. Sets the account status to 1 (Open).
16	Updates the PBF record. If the update fails, then the transaction is denied with response code 062.
17	If the EXP DATE CHECK IND field in the CPF is set to 0, the expiration date is not checked. The date is open ended. If the EXP DATE field in the CPF is not 0 and the DURATION field is not 0, the EXP DATE is calculated as today's date plus DURATION. If the CPF DURATION field is set to 0, then the CPF EXP DATE field value is used. If the EXP DATE CHECK field is not 0 and the CPF DURATION field is 0 and the CPF EXP DATE field is 0000, the transaction is denied. Updates the PBF usage accumulator fields.
18	If the transaction is approved, the module sets the CAF card status to a value of 1 (Open).

Additional Card Activation Transaction

When an additional card activation transaction is processed, the Router/Authorization module performs the following steps:

Steps	Processing
1	Verifies that the CAF Card Type value is valid for Stored Value transactions. Valid values are: SR = Stored Value reloadable SN = Stored Value non-reloadable If the card type is not valid, the transaction is denied with response code 055.
2	Retrieves the account index. If the account index is not found, the transaction is denied with response code 060.
3	Stores the account information from the existing CAF record.
4	Retrieves the Stored Value token from the PSTM. If the token is not available for any reason, the module declines the transaction with response code 074.
5	Unlocks the existing card's CAF record.
6	Reads the CAF record associated with the data found in the token. If the CAF record is not found, a new CAF is added. If the read fails for any other reason, the module declines the transaction with response code 206.
7	Verifies that the CAF card status is 0 (Issued Not Active). If the value is not 0, the transaction is denied with response code 058.
8	Updates the account information in the CAF record with the information saved from the existing card's CAF record.
9	Uses the expiration date from the existing CAF record for the additional CAF record.
10	Sets the CAF card status to a value of 1 (open).

Replenishment Transaction

When an replenishment transaction is processed, the Router/Authorization module performs the following steps:

Steps	Processing
1	Verifies that the CAF Card Type value is valid for Stored Value transactions. Valid values are: SR = Stored Value reloadable SN = Stored Value non-reloadable If the card type is not valid, the transaction is denied with response code 055.
2	Verifies the amount requested. The amount must be in the range defined by the CPF minimum load limit and the maximum load limit. If the amount is over the maximum load limit, the transaction is denied with response code 092.
3	Verifies that the amount requested does not raise the ledger balance over the limit. The new balance must be less than the maximum card balance in the CPF. If the amount is over the limit, the transaction is denied with response code 095.
4	Verifies that the number of refunds and replenishments limit has not been exceeded. If exceeded, the transaction is declined with response code 087.
5	Verifies that the per refund limit will not be exceeded. If exceeded, the transaction is denied with response code 081.
6	Verifies that the offline refund limit will not be exceeded. If exceeded, the transaction is denied with response code 080.
7	Verifies that the total refund limit will not be exceeded. If exceeded, the transaction is denied with response code 083.
8	Retrieves the account index. If no index is found, the transaction is denied with response code 060.
9	Reads the PBF record associated with the CAF record. If unable to read the PBF, the transaction is denied with response code 061.

Steps**Processing**

- 10 Updates the PBF record. If the update fails, the transaction is denied with response code 062. Updates the CAF usage accumulator fields.

Full Redemption Transaction

When a full redemption transaction is executed, the Router/Authorization module performs the following steps:

Steps	Processing
1	Determines if there are any pre-authorization holds. If there are, the transaction is denied with response code 074.
2	Retrieves the account index. If no index is found, the transaction is declined with response code 060.
3	Reads the PBF record associated with the CAF record. If unable to read the PBF, the transaction is declined with response code 061.
4	If the ledger balance in the PBF is greater than the max balance as cash amount specified in the CPF, the transaction is declined with response code 095.
5	If the ledger balance is below the limit, then the ledger balance is moved to the AMT1 field in the PSTM. The module updates the usage accumulator fields in PBF. The ledger balance and available balance are set to zero after subtracting AMT1.
6	If the ledger balance is below the limit, the ledger balance is moved to the AMT1 field in the PSTM. Updates the usage accumulator fields in PBF. The ledger balance and available balance are set to zero after subtracting AMT1.
7	Updates the CAF usage accumulators.

Offline Stored Value Transactions

Stored value cards cannot be activated, replenished or redeemed in an offline mode at the POS device. If an acquirer wants to support these transactions because of customer service considerations, the terminal/controller can perform stand-in authorization and then send the transaction to BASE24 as a 0200 request message. These transactions must include an S in the message subtype field. The terminal/controller then receives a 0210 response message from BASE24.

If the transaction is later denied when BASE24 is online, the terminal/controller should treat the transaction as an exception and not include it in the terminal totals. The stored value update transaction is not made to the account and the acquirer will have to handle this situation as a customer service issue.

ACI Worldwide Inc.

4: SPDH Message Formats

The BASE24-pos Standard Device Handler (SPDH) module is used to communicate with POS devices supporting the SPDH message format. The SPDH allows the retailer or processor to support all standard POS transactions as well as the following Stored Value transactions online:

- Card activation T
- Additional card activation
- Replenishment
- Full redemption

For information about the functionality of the SPDH Device Handler module, see the *BASE24-pos Standard POS Device Configuration Manual*.

Transaction Codes

The SPDH module determines the type of BASE24-pos transaction associated with a message using two fields in the standard header. These two fields are the Message Type and Transaction Code. The Message Type field specifies the transaction as being either a financial transaction (F) or an administrative transaction (A). The Transaction Code field identifies the type of transaction, as determined by the terminal.

The following table describes how SPDH transaction codes are mapped to BASE24-pos transaction codes:

Message Type	SPDH Transaction Code	Transaction Description	BASE24-pos Transaction Code
F	15	Card Activation	25
F	16	Additional Card Activation	26
F	17	Replenishment	27
F	18	Full Redemption	28

Terminal Messages

The POS transaction messages transmitted from an SPDH-compliant supported terminal to BASE24-pos must support the Stored Value transaction set. These transaction messages are described below and on the following pages.

Card Activation Transaction

The card activation transaction is initiated when a customer desires to purchase a stored value card.

When the SPDH module receives the transaction, it builds a PSTM Card Activation request.

The Card Activation responses received by the SPDH module are formatted into Card Activation responses and returned back to the device.

Additional Card Activation Transaction

The additional card activation transaction is initiated when a customer desires to purchase an additional card to tie to an existing stored value card account.

This transaction includes the additional card Track data in SFID 5 of FID 6.

When the SPDH module receives the transaction, it builds a PSTM Additional Card Activation request.

The responses that are received by the SPDH are formatted into Additional Card Activation responses and returned back to the device.

Replenishment Transaction

The replenishment transaction is initiated when a customer desires to add funds to an activated stored value card. This transaction causes the balance of the cards associated with a PBF record to be increased by the replenishment amount.

When the SPDH module receives the transaction, it builds the PSTM Replenishment request.

The Replenishment SPDH responses received by the SPDH are formatted into Replenishment responses and returned back to the device.

Full Redemption Transaction

The full redemption transaction is initiated when a customer desires to remove the remaining value from the stored value card.

This transaction causes the balance of the cards associated with a PBF record to be reduced to zero.

When the SPDH module receives the transaction, it builds the PSTM Full Redemption request.

The Full Redemption SPDH responses received by the SPDH are formatted into Full Redemption responses and returned back to the device.

Card Activation Transaction Request

The following table describes the fields in the standard message header and the optional data fields in a sample card activation transaction request.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (Dial or leased line terminal, link type switched)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (employee ID is validated)
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (Card activations are financial transactions)
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	15 (Card activation transaction)
PROCESSING FLAG 1	1 (Respond and do not disconnect)
PROCESSING FLAG 2	0 (Processing Flag 2 is not applicable in requests)
PROCESSING FLAG 3	0 (All employee totals for a specific terminal)
RESPONSE CODE	000 (Response code is not applicable in requests)

Field	Description
Optional Data Fields	
FID B AMOUNT 1	Amount to be loaded into the new account. This amount is placed in the TRAN.AMT-1 field of the PSTM.
FID q TRACK 2/CUSTOMER	Contains the track 2 of the card being activated.
FID 6 SFID S Stored Value Data	Contains the balance as cash flag, card balance and expiration date. This data is placed in the Stored Value token.

Card Activation Transaction Response

The following table describes the fields in the standard message header and the optional data fields in a sample card activation transaction response.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (echoed from the request)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056 (echoed from the request)
EMPLOYEE ID	987654 (echoed from the request))
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (echoed from the request)
MESSAGE SUBTYPE	O (echoed from the request)
TRANSACTION CODE	15 (echoed from the request)
PROCESSING FLAG 1	1 (Respond and disconnect)
PROCESSING FLAG 2	0 (No download waiting)
PROCESSING FLAG 3	0 (Processing flag 3 is not applicable in this type of response)
RESPONSE CODE	001(Response code is not applicable in requests).
Optional Data Fields	
FID B AMOUNT 1	Amount to be loaded into the new account.
FID q TRACK 2/CUSTOMER	Contains the Track 2 data of the card being activated.

Field	Description
FID 6 Stored Value Data	Contains the balance as cash flag, card balance, and expiration date.

Additional Card Activation Transaction Request

The following table describes the fields in the standard message header and the optional data fields in a sample additional card activation transaction request.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (Dial or leased line terminal, link type switched)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (employee ID is validated)
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (Additional card activations are financial transaction).
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	16 (Additional card activation transaction)
PROCESSING FLAG 1	1 (Respond and do not disconnect)
PROCESSING FLAG 2	0 (Processing flag 2 is not applicable in requests)
PROCESSING FLAG 3	0 (All employee totals for a specific terminal)
RESPONSE CODE	000 (Response code is not applicable in requests).

Field	Description
Optional Data Fields	
FID q TRACK 2/CUSTOMER	Contains the Track 2 data for the primary card, which has been previously activated using a card activation request. This data is placed in the Track 2 field of the PSTM.
FID 6 SFID S Stored Value Data	Contains the additional card Track 2 data in Additional Card Activation requests. This data is placed in the Stored Value token.

Additional Card Activation Transaction Response

The following table describes the fields in the standard message header and the optional data fields in a sample Additional Card Activation transaction response.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (Dial or leased line terminal, link type switched)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (echoed from request)
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (echoed from the request)
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	16 (echoed from the request)
PROCESSING FLAG 1	1 (Respond and disconnect)
PROCESSING FLAG 2	0 (No download waiting)
PROCESSING FLAG 3	0 (Processing flag 3 is not applicable in this type of response)
RESPONSE CODE	001 (Approved)
Optional Data Fields	
FID q TRACK 2/CUSTOMER	Contains the Track 2 data for the primary card, which has been previously activated using a Card Activation request.
FID 6 SFID S Stored Value Data	Contains the additional card Track 2 data.

Replenishment Transaction Request

The following table describes the fields in the standard message header and the optional data fields in a sample Replenishment transaction request.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (Dial or leased line terminal, link type switched)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (employee ID is validated)
CURRENT DATE	The current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (Replenishment transactions are financial transactions).
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	17 (Replenishment transactions)
PROCESSING FLAG 1	1 (Respond and do not disconnect)
PROCESSING FLAG 2	0 (Processing flag 2 is not applicable in requests)
PROCESSING FLAG 3	0 (All employee totals for a specific terminal)
RESPONSE CODE	000 (Response code is not applicable in requests).

Field	Description
Optional Data Fields	
FID B AMOUNT1	Contains the replenishment amount. This amount is placed in the TRAN.AMT-1 field of the PSTM.
FID q TRACK 2/CUSTOMER	Contains the Track 2 data for the primary card or the secondary card associated with the account being replenished. This data is placed in the TRACK2 field of the PSTM.

Replenishment Transaction Response

The following table describes the fields in the standard message header and the optional data fields in a sample Replenishment Transaction response.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (echoed from the request)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (echoed from request)
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (echoed from the request)
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	17 (echoed from the request)
PROCESSING FLAG 1	1 (Respond and disconnect)
PROCESSING FLAG 2	0 (No download waiting)
PROCESSING FLAG 3	0 (Processing flag 3 is not applicable in this type of response)
RESPONSE CODE	001 (Approved)
Optional Data Fields	
FID B AMOUNT 1	Contains the replenishment amount.
FID q TRACK 2/CUSTOMER	Contains the Track 2 data of the primary card or the secondary card associated with the account being replenished.

Field	Description
FID 6 SFID S Stored Value Data	Contains the new card balance and expiration date.

Full Redemption Transaction Request

The following table describes the fields in the standard message header and the optional data fields in a sample Full Redemption transaction request.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (Dial or leased line terminal, link type switched)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (employee ID is validated)
CURRENT DATE	The current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (Full Redemption transactions are financial transactions)
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	18 (Replenishment transactions)
PROCESSING FLAG 1	1 (Respond and do not disconnect)
PROCESSING FLAG 2	0 (Processing flag 2 is not applicable in requests)
PROCESSING FLAG 3	0 (All employee totals for a specific terminal)
RESPONSE CODE	000 (Response code is not applicable in requests)

Field	Description
Optional Data Fields	
FID q TRACK 2/CUSTOMER	Contains the Track 2 data of the primary card or the secondary card associated with the account being replenished. This data is placed in the TRACK2 field of the PSTM.

Full Redemption Transaction Response

The following table describes the fields in the standard message header and the optional data fields in a sample Full Redemption Transaction response.

Field	Description
Message Header Fields	
DEVICE TYPE	9. (echoed from the request)
TRANSMISSION NUMBER	If set to 00, the transmission number is not checked.
TERMINAL ID	POS-TERMINAL-056
EMPLOYEE ID	987654 (employee ID is validated)
CURRENT DATE	Current date is optional. If used, the CURRENT TIME must be filled in.
CURRENT TIME	Current time is optional.
MESSAGE TYPE	F (echoed from the request)
MESSAGE SUBTYPE	O (Online)
TRANSACTION CODE	18 (echoed from the request)
PROCESSING FLAG 1	1 (Respond and disconnect)
PROCESSING FLAG 2	0 (No download waiting)
PROCESSING FLAG 3	0 (Processing flag 3 is not applicable in this type of response)
RESPONSE CODE	001 (Approved)
Optional Data Fields	
FID q TRACK 2/CUSTOMER	Contains the Track 2 data of either the primary or secondary card associated with the account being redeemed.
FID 6 SFID 6 Stored Value Data	Contains the Stored Value data information, including the remaining balance to be redeemed.

5: ISO Host Message Formats

The BASE24-pos ISO Host Interface process is responsible for the online message transfer between a BASE24 system and a host platform. In order to support stored value transactions, the acquirer's POS host must be modified to accept these transactions and pass them to BASE24 for authorization.

For information about the functionality of the ISO Host Interface process, see the *BASE24 ISO Host Interface Manual*.

Transaction Codes

The following table describes the four Stored Value transactions that are processed through the BASE24-pos Host Interface process.

BASE24-pos Transaction Code	HISO Transaction Code	Transaction Description
25	98	Card Activation
26	99	Additional Card Activation
27	28	Replenishment
28	17	Full Redemption

Terminal Messages

The following POS transaction messages transmitted from a host to BASE24-pos must support the Stored Value transaction set.

Card Activation Transaction

The card activation transaction is initiated when a customer desires to purchase a stored value card.

When the BASE24-pos ISO Host Interface module receives the transaction from a host, it builds a BASE24-pos Standard Internal Message (PSTM) Card Activation request.

The Card Activation responses received by the BASE24-pos ISO Host Interface module are formatted into Card Activation responses and returned to the host.

Additional Card Activation Transaction

The additional card activation transaction is initiated when a customer desires to purchase an additional card to tie to an existing stored value card account.

This transaction includes the additional card Track data in the Stored Value token.

When the BASE24-pos ISO Host Interface module receives the transaction from the host, it builds a PSTM Additional Card Activation request.

The responses that are received by the BASE24-pos ISO Host Interface module are formatted into Additional Card Activation responses and returned to the host.

Replenishment Transaction

The replenishment transaction is initiated when a customer desires to add funds to an already activated stored value card. This transaction causes the balance of the cards associated with a PBF record to be increased by the replenishment amount.

When the BASE24-pos ISO Host Interface module receives the transaction from the host, it builds the PSTM Replenishment request.

The Replenishment responses received by the BASE24-pos ISO Host Interface module are formatted into Replenishment responses and returned to the host.

Full Redemption Transaction

The full redemption transaction is initiated when a customer desires to remove the remaining value from the stored value card.

This transaction causes the balance of the cards associated with a Positive Balance File (PBF) record to be reduced to zero.

When the BASE24-pos ISO Host Interface module receives the transaction from the host, it builds the PSTM Full Redemption request.

The Full Redemption responses received by the BASE24-pos ISO Host Interface module are formatted into Full Redemption responses and returned to the host.

6: Batch Processing and Reporting

This section describes the following batch processes used by the BASE24-pos Stored Value add-on product:

- History File Cleanup (HFCLUP)
- Stored Value History File Extract (SVHFX)
- CAF/PBF Cleanup (CARDCLN)

This section also describes the following BASE24-pos reports that support Stored Value transaction information:

- Daily Network Settlement Report (B24POS01)
- Daily Cardholder Activity Report (B24POS02)
- Daily Clerk Totals Report (B24POS04)
- Daily Retailer Summary Report (B24POS11)
- Daily Retailer Reconciliation Report (B24POS35)
- Periodic Institution Summary Report (B24POS21)
- Periodic Retailer Summary Report (B24POS22)

History File Cleanup Module (HFCLUP)

The HFCLUP module deletes old records from the Stored Value History File (SVHF). This stand-alone module is executed from a TACL prompt. Parameters are passed in the TACL prompt specifying the specific SVHF partition to clean up.

The process determines the beginning key values of the partitions programmatically. The records are then deleted when they are older than the retention period specified in the POS-SVHF-RET-PERIOD LCONF param. Each HFCLUP module is started from a TACL prompt with a run file and is associated with one partition of the SVHF file. When the process is started, the assigns and params are passed to the process.

Program Startup

The History File Cleanup program is started when the HFCLUPR file is executed. This file resides on the PS $_{xx}$ SV subvolume, where XX is the number of the current release.

HFCLUPR File Example

```
CLEAR ALL PARAM
CLEAR ALL ASSIGN
PARAM PART PART##
ASSIGN LCONF, $VOL.SUBVOL.LCONF
RUN HFCLUP/NAME $HCLN, CPU XX, PRI YYY, NOWAIT, OUT $S.#HCLUP/
```

Syntax

HFCLUP — The name of the History File Cleanup executable.

— The name of the partition number (valid ranges are part01 to part15).

xx — The CPU in which the History File Cleanup program will run.

yyy — Specifies the priority at which the History File Cleanup process will run.

OUT — The spooler location where the output report will be generated.

Reports

The HFCLUP module generates a report at the location specified in the startup message (e.g., \$S.#HCLUP). The report contains the following information:

- Date and time (start and end) of the run
- The number of records deleted
- The primary file name
- Indicates whether the target file was the primary partition, or a secondary partition with the partition number
- Any critical messages generated during the run

HFCLUP Processing Steps

When the HFCLUP program is executed, it performs the following steps:

Steps	Processing
1	Reads the startup message and retrieves assigns, params, and system messages. The process opens the out file (for the processing report) specified in the startup message and the primary collector (for logging event messages).
2	Verifies that all required TACL assigns and params are present. If any of the required assigns or params are missing, the process logs an informative message and terminates.
3	Opens the LCONF specified in the TACL assign and reads the appropriate assigns and params. If any of the required assigns or params are missing, the process logs an informative message and terminates.
4	Determines whether the file being processed is a primary or secondary partition from the TACL Param PART. The process programmatically determines the beginning key value for the file and partition being processed and the beginning key value for the next partition (if one exists) from the file specified in the LCONF assign POS-SVHF. If the file being processed is not partitioned, or if the file being processed is the last secondary partition (i.e., PART15), there is no next partition and a read returning an EOF (error = 1) indicates there are no further records to process. In all other cases, comparing the key value of the current record with the beginning key value of the next partition is used to determine when processing has finished.
5	Opens the SVHF for read/write access and shared mode. If any error occurs during the open, the process will log an informative message and terminate.
6	After successful initialization, determines the current date and calculates the transaction posting date for which a given record has passed the record retention period (retention date = current date – POS-SVHF-RET-PERIOD).

Steps	Processing
7	Begins processing records in the designated SVHF. The process reads records from the SVHF partition in sequential order from the beginning of the file (either the primary or one of the secondary partitions if they exist).
8	If the record key value is less than the ending key value, the transaction posting date field (TRAN-DAT-TIM) is compared with the record retention date.
9	If the record has expired and is no longer in the record retention period ($\text{TRAN-DAT-TIM} < \text{current date} - \text{POS-SVHF-RET-PERIOD}$), then the record is deleted from the SVHF. If the record has not expired, then the next record is read.
10	When the process is reading records from the last secondary partition, an error is returned ($\text{error} = 1$) indicating the last record in the file has already been processed.
11	When either the last record (from the associated SVHF partition) has been processed, or an end-of-file (EOF) has been detected, the process generates a report to the designated out file.
12	The process then closes the SVHF and primary collector, logs an informative message, and terminates.

Stored Value History File Extract Module (SVHFX)

The Stored Value History File Extract (SVHFX) module extracts the Stored Value History File (SVHF) to tape or disk. This module can be run automatically or programatically.

Standard BASE24 extract processing allows automatic extracting. This feature is configured in the EXTRACT TIME field on screen 1 of the Extract Configuration File (ECF).

Configurable fields in the ECF allow the extract of the SVHF records whose log timestamps fall within the starting and ending dates and times specified by the user.

Placing a value of all zeros in the Start Date, Time, and End Date fields on the ECF screen 7 causes an automatic SVHF extract for the previous calendar day.

The *BASE24 Refresh and Extract Operators Manual* describes extract processing in detail.

CAF/PBF Cleanup Program (CARDCLN)

The CAF/PBF Card Cleanup Program deletes CAF and PBF records for CAF records that have expired dates and PBF records with zero balances. This program is a standalone process that deletes expired records from the CAF and PBF. It is executed from a TACL prompt with parameters passed to specify the particular partition to clean up. The criteria for deleting these records are:

- The CAF expiration date no longer falls within the retention period specified in the LCONF.
- The CAF card type is “Stored Value”.
- The PBF balance is 0 or the PBF read fails with an end of file error.
- The CAF expiration date is not equal to 0000. A value of 0000 indicates that the Stored Value card does not expire.

If any of the above criteria have not been met, neither record will be deleted.

Process Startup

The CAF/PBF Cleanup program is started when the CARDCLNR file is executed. This run file resides on the PS xx SV subvolume where xx is the number of the current release.

CARDCLNR File Example

```
CLEAR ALL PARAM
CLEAR ALL ASSIGN
ASSIGN LCONF, $VOL.SUBVOL.LCONF
RUN CARBDLN/NAME $CCLN, CPU XX, PRI YYY, NOWAIT, OUT $.#CCLUP/
```

Syntax

CARDCLN — The name of the CAF/PBF Cleanup program executable.

NAME — The Card Cleanup process name.

xx — The CPU the program should be run in.

yyy — The startup priority of the program

OUT — The spooler location where the output report will be generated.

Reports

The CARDCLN process generates a report at the location specified in the startup message (e.g., \$s.#CCLUP). The report contains the following information:

- Date and time (start and end) of the run
- The number of records deleted and the associated file names (CAF and PBF)
- The primary file name
- Any critical messages generated during the associated run

CAF/PBF Cleanup (CARDCLN) Processing Steps

When the CARDCLN program is executed, it performs the following steps:

Steps	Processing
1	Reads the startup message and retrieves assigns, params, and system messages. The process opens the out file (for the processing report location) specified in the startup messages and the primary collector (for logging event messages).
2	Verifies that all required TACL assigns and params specified above are present. If any of the required assigns or params are missing, the process logs an informative message and terminates.
3	Opens the LCONF specified in the TACL assign and reads the appropriate assigns and params. If any of the required assigns or params are missing, the process logs an informative message and terminates.
4	Determines whether the file being processed is a primary or secondary partition from the TACL param PART. The process programmatically determines the beginning key value for the file and partition being processed and the beginning key value for the next partition (if one exists) from the file specified in the LCONF assign POS-SVHF. If the file being processed is not partitioned, or if the file being processed is the last secondary partition (i.e., PART15), there is no next partition and a read returning an EOF (error = 1) indicates there are no further records to process. In all other cases, comparing the key value of the current record with the beginning key value of the next partition is used to determine when processing has finished.
5	Deletes expired records from the CAF and PBF files based on the following criteria: • CAF is expired and the expiration date no longer falls within the LCONF param • CAF Card Type is “Stored Value” • PBF balance is 0 or PBF read fails with an EOF error • CAF expiration date is not equal to 0000

BASE24-pos Reports

The following BASE24-pos reports include BASE24-pos Stored Value transactions:

- Daily Network Settlement Report (B24POS01)
- Daily Cardholder Activity Report (B24POS02)
- Daily Clerk Totals Report (B24POS04)
- Daily Retailer Summary Report (B24POS11)
- Daily Retailer Reconciliation Report (B24POS35)
- Periodic Institution Summary Report (B24POS21)
- Periodic Retailer Summary Report (B24POS22)

For detailed information on these BASE24-pos reports, refer to the ***BASE24-pos Settlement and Reporting Manual***.

7: Dormancy and Escheatment

BASE24 customers that support stored value cards have the ability to apply dormancy fees to cards that have not expired but have not been used for a period of time. The card issuer has the ability to collect fees and eventually take the account balance down to zero so that the card can be cleared from the system.

Escheatment reports associated with stored value cards allow an issuer to turn over inactive account information as dormant unclaimed assets.

This section describes the processing that is common to both dormancy processing and escheatment reporting. Also included are examples of dormancy and escheatment reports.

Dormancy Processing and Escheatment Reporting

The dormancy process and escheatment reporting process are one process. The following describes the processing that is common to both dormancy processing and escheatment reporting. Where processing differs, it is noted.

Initialization Steps

The following are initialization steps for dormancy processing and escheatment reporting.

1. Reads the startup message from the TACL prompt.
2. Stores the LCONF file name that was included in the TACL startup message. The fully qualified file name enables this process to access the LCONF for the assigns and params used in processing.
3. Stores the processing option that was included in the TACL startup message.
 - a. If the processing option is D, dormancy processing is performed.
 - b. If the processing option is E, escheatment processing is performed.
 - c. If the processing option is not D or E, default to dormancy processing and set the processing option field to D.
4. Stores the institution ID option that was included in the TACL startup message. The institution ID option can specify a particular financial institution or all institutions (i.e., using four asterisks).
5. Stores the error report location included in the OUT option of the TACL startup message, for example, Out \$.#errrpt.
6. Retrieves the necessary assigns and params from the LCONF.
7. Opens the primary collector for event messages.
 - a. If the open fails on the primary collector, open the alternate collector.
 - b. If the open fails on the alternate collector, call netabend.
8. Determines whether processing of the Stored Value Dormancy File (SVDF) is required?
 - a. If the processing option indicates that dormancy processing is required, opens the SVDF, allocates extended memory space, and builds the SVDF extended memory table.
 - b. Otherwise, skip SVDF processing.

9. Determines whether processing of the POS Transaction Log File (PTLF) is required?
 - a. If the processing option indicates that dormancy processing is required, every record that is written to the SVDF is also written to the PTLF with a PTLF record type of 22. The Subtype token is added with the subtype set to C003DORMANCY TRANSACTION.
 - b. If the processing option indicates that escheatment processing is required, every record that is written to the SVDF is also written to the PTLF with a PTLF record type of 22. The Subtype token is added with the subtype set to C004ESCHEATMENT TRANSACTION.
 - c. Otherwise, skip PTLF processing.
10. Opens the Card Prefix File (CPF), allocates extended memory space, and builds the CPF extended memory tables. This is a flat segment.
 - a. If the processing option indicates dormancy processing, perform the following steps for building the CPF extended memory table:
 - 1) If the FIID in the CPF does not equal the institution ID option, and the institution ID option does not equal four asterisks, do not move the CPF record data into extended memory.
 - 2) If the CPF card type is one of the card types specified as a stored value card type, determine whether any dormancy fees are possible by interrogating the SVDF records.
 - 3) Compare the Retailer ID and Card Type fields in the CPF against the corresponding fields in the SVDF using the following search hierarchy:

Retailer ID	Card Type
Exact Match	Exact Match
Exact Match	Wildcard with asterisks
Wildcard with asterisks	Exact Match
Wildcard with asterisks	Wildcard with asterisks

- 4) If a match is found, move the appropriate data from the CPF record into extended memory, move the match that was found against the SVDF, and move the FIID of the base segment of the CPF into the FIID internal table if it does not already exist in the FIID internal table.
- 5) Otherwise, do not move the CPF data into extended memory.

- b. If the processing option indicates escheatment processing, perform the following for building the CPF extended memory table:
 - 1) If the FIID does not equal the institution ID option, and the institution ID option does not equal four asterisks, do not move the CPF data into extended memory.
 - 2) If the CPF escheatment flag is not set to Y, do not move the CPF data into extended memory.
 - 3) If the CPF card type equals one of the stored value card types, determine whether escheatment processing is possible by interrogating the escheatment internal table.
 - 4) If a match is found, move the appropriate data from the CPF record into extended memory, and move the FIID of the base segment of the CPF into the FIID extended memory table if it does not already exist in the FIID internal table.
 - 5) Otherwise, do not move the CPF data into extended memory.
- 11. Allocates extended memory space for the total fees per Retailer ID, and builds the Retailer ID extended memory tables.
 - a. Search the Retailer ID extended memory table for the Retailer ID retrieved from the CPF extended memory table before creating another entry in the Retailer ID extended memory segment.
 - b. Move each stored value Retailer ID found in the CPF extended memory table to this table if it does not already exist.
 - c. Set the FIID for each stored value Retailer ID to the associated value in the CPF extended memory table.
 - d. Set the total fee bucket per Retailer ID to zero.
 - e. Set the total fee bucket per stored value card type to zero.
- 12. Opens the Institution Definition File (IDF), allocates extended memory space, and builds the IDF extended memory tables.
 - a. If the FIID in the IDF equals one of the FIIDs specified in the FIID internal table, perform the following tasks:
 - 1) Move the appropriate data from the IDF record into extended memory.
 - 2) If the PBF4 file name in the IDF record is not spaces, move this value into the stored value PBF file name field for this financial institution.

- 3) If the PBF4 file name in the IDF record is spaces, move the PBF1 file name into the stored value PBF file name field for this financial institution.
 - b. Otherwise, do not move the IDF data into extended memory.
13. Set a global timestamp to the current date and time for the start of the report.

Account Processing

The following steps describe dormancy processing and escheatment reporting for a particular account.

1. Opens the Cardholder Authorization File (CAF) specified in the Institution Definition File (IDF) extended memory segment for this FIID.
2. Retrieves the CAF record.

This begins a loop where all CAF records must be read and locked for this FIID to determine which Positive Balance File (PBF) accounts are dormant and must be charged a fee.
3. Determines whether the FIID for this CAF record matches the FIID or the wildcard in the startup message. If there is a match, processing continues. If there is no match, unlock the CAF and read the next record.
4. Determines whether the card type in this CAF record equals one of the valid stored value card types. If a match is not found, unlock this CAF record and retrieve the next CAF record by continuing with step 2.

Check the Ignore Card Status Type table. If our card type is found, do not apply dormancy fees to this card.
5. Searches the CPF extended memory segment using the PAN from the CAF record and the calculated length of the actual PAN, not the length of the PAN field. A match is found based on the best match on the prefix, prefix length, and PAN length. If a match is not found, generate an event, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
6. Determines whether a fee record exists for this CPF record. This was originally determined at the time of initialization.
7. Retrieves the PBF record using the first account in the account segment of the CAF.
 - a. The PBF file name is contained in the stored value PBF file name field in the IDF extended memory table.

- b. The key information to read the PBF record is obtained from the following locations:
 - 1) The FIID from the base segment of the CAF.
 - 2) The account type from the account segment of the CAF.
 - 3) The account number from the account segment of the CAF.
 - c. If the PBF record cannot be retrieved, generate an event, unlock this CAF record and retrieve the next CAF record by continuing with step 2.
8. Checks the PBF.POS.SV_RPT_TS field against the global timestamp for the start of the report if they are not equal, then processes this PBF if it has not been processed yet.
9. Determines whether there is a fee for this particular PBF record.
- a. If the processing option indicates dormancy processing, perform the following steps:
 - 1) Find the SVDF record in the SVDF extended memory segment using the following:

Match retailer ID from the CPF extended memory segment.

Match card type from the CPF extended memory segment.

Stored value activation state field in the PBF record.

A match was already found on the retailer ID and card type based on the CPF record. However, the activation state that is also part of the primary key of a SVDF record, so another search must be performed against the SVDF extended memory segment.
 - 2) Compare the Match Retailer ID, Match Card Type, and Activation State fields against the corresponding fields in the SVDF using the following search hierarchy:

Match Retail ID	Match Card Type	Stored Value Activation State
Exact Match	Exact Match	Exact Match
Exact Match	Exact Match	Wildcard with asterisks

- 3) A match was already found on the retailer ID and card type based on the CPF record, and, therefore, the activation state that is also part of the primary key into the SVDF record must be used in the search.

- 4) If a match cannot be found, unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
 - b. If the processing option indicates escheatment processing, perform the following steps:
 - 1) Find the escheatment internal table entry using the following:
Stored value activation state field in the PBF record. If this is not set, then continue with step 2.

Financial institution state from the IDF extended memory segment.
 - 2) Compare the stored value activation state field against the corresponding field in the PBF (or possibly the IDF extended memory segment).

If this PBF is not spaces, compare the stored value activation state field in the CPF against the value in the PBF.

If this PBF field contains spaces, compare the stored value activation state to the financial institution state from the IDF extended memory segment.
 - 3) If a match cannot be found, unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
10. Determines whether this PBF account is dormant.

For dormancy processing, do not assess a fee on an account with a zero PBF balance.

For escheatment processing, do not escheat an account with a zero PBF balance.
 - a. Determine which days dormant value to use based on the processing option value.
 - 1) If the processing option indicates dormancy processing, use the days dormant value from the SVDF record.
 - 2) If the processing option indicates escheatment processing, use the days dormant value from the escheatment internal table.

- b. Add the obtained days dormant date value to either the PBF record's last deposit date or the PBF record's last withdrawal date, depending on the field that is greater. This calculation yields the expiration date (or dormancy date).
 - 1) If both the last deposit date and the last withdrawal date fields in the PBF record equal zeroes or spaces, this record has not been used at all, and, therefore, may just be a newly-activated card. Unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
 - 2) Otherwise, continue on with the following step where the dates are compared.
 - c. Compare the dormancy date against the today's calendar date.
 - d. If the dormancy date is less than or equal to today's calendar date, the account is dormant.
 - e. Otherwise, unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
11. Determines whether this account has been assessed a dormancy fee within the dormancy period, if the processing option indicates dormancy processing.
- a. Add the last transaction timestamp of the PBF record (i.e., only the year, month, and day) to the period field in the SVDF record.
 - b. Compare the last transaction timestamp date against today's calendar date.
 - c. If the last transaction timestamp is less than or equal to today's calendar date, the account is dormant.
 - d. Otherwise, unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
12. If the balance prior to dormancy fee field in the PBF is zero (the first time a dormancy fee has been applied), sets the balance prior to dormancy fee field in the PBF record to the available balance in the PBF record, if the processing option indicates dormancy processing.
13. If the date prior to dormancy fee field in the PBF is zeros or spaces (first time a dormancy fee has been applied), sets the date prior to dormancy fee field for the PBF record to the current calendar date when the following conditions are met:
- a. Processing option indicates dormancy processing.
 - b. Balance prior to dormancy fee field in the PBF record does not equal zero.

14. Calculates the dormancy fee if the processing option indicates dormancy processing.
 - a. The dormancy fee is determined by using the following values from the database:
 - 1) The available balance from the PBF record is the transaction amount.
 - 2) The flat fee from the SVDF record.
 - 3) The percent fee from the SVDF record.
 - b. If the flat fee does not equal zero, set the dormancy fee equal to the smaller of the two values:
 - 1) Flat fee
 - 2) Balance prior to dormancy fee
 - c. If the flat fee equals zero, set the dormancy fee equal to the smaller of the two values:
 - 1) The calculated value of multiplying the balance prior to dormancy fee field by the percent fee, and then dividing by 100.
 - 2) The balance prior to dormancy fee field.
15. Recalculates the last transaction timestamp in the PBF record if the processing option indicates dormancy processing.
16. If the processing option indicates dormancy processing, subtracts the calculated dormancy fee from the available balance field in the PBF record. Makes sure that the PBF available balances does not go negative if the dormancy fee is greater than the available balance, sets the dormancy fee to the available balance and sets the available balance to zero.
17. If the processing option indicates dormancy processing, subtracts the calculated dormancy fee (calculated above) from the ledger balance field in the PBF record.
18. Sets the PBF.POS.SV_RPT_TS field to the global report start timestamp.
19. If the processing option indicates escheatment processing, sets the write length for the PBF record to zero.
20. Writes the PBF back to disk.
 - a. If the processing option indicates escheatment processing, write the report lines to the escheatment report. These report lines include the following information:
 - 1) The PAN from the CAF record.

- 2) The balance prior to dormancy fee field from the PBF record.
 - 3) The calculated value determined by subtracting the available balance from the PBF record from the balance prior to dormancy fee from the PBF record.
 - 4) The available balance from the PBF record.
 - 5) The stored value retailer ID from the entry in the CPF extended memory segment.
 - 6) The stored value card type from the entry in the CPF extended memory segment.
 - 7) The date prior to dormancy fee from the PBF record.
 - 8) The last deposit date or the last withdrawal date in the PBF, whichever is greater.
 - 9) The stored value activation state in the PBF or the financial institution state in the IDF extended memory depending on whether the stored value activation state in the PBF contains spaces.
 - b. If the write fails, generate an event, unlock this PBF record, unlock this CAF record, and retrieve the next CAF record by continuing with step 2.
21. If the processing option indicates escheatment process, deletes the CAF record.
- a. If the write fails, generate an event.
22. Writes the appropriate information the SVHF.
- a. Set the SVHF.MSG_TYP field to D for dormancy or E for escheatment.
 - b. If the write fails, generate events with the information should have been written to the SVHF, unlock this CAF record (if dormancy processing), and continue processing.
23. If the processing option indicates dormancy processing, updates the dormancy report totals.
- a. Add the dormancy fee to the total fee for the FIID. This total is located in the IDF extended memory table.
 - b. Add the dormancy fee to the total fee for the appropriate retailer ID. This total is located in the retailer ID extended memory table.
 - c. Add the dormancy fee to the total fee for the appropriate stored value card type. This total is located in the retailer ID extended memory table.

24. If the processing option indicates escheatment processing, updates the escheatment report totals.
 - a. Add the available balance to the total fee for the FIID. This total is located in the IDF extended memory table.
 - b. Add the available balance to the total fee for the appropriate retailer ID. This total is located in the retailer ID extended memory table.
25. Performs the following processing for an institution after all CAF records for that institution are read.
 - a. If the processing option indicates dormancy processing, write the total fees to the dormancy report for this institution.
 - 1) The total fee for the FIID from the IDF extended memory table is written to the report.
 - 2) The total fees for each retailer ID located in the retailer ID extended memory table are added together and written to the report.
 - 3) The total fees for each stored value card type located in the retailer ID extended memory table are added together and written to the report.
 - b. If the processing option indicates escheatment processing, writes the total fees to the escheatment report for this institution.
 - 1) The total fee for the FIID located in the IDF extended memory table is written to the report.
 - 2) The total for each retailer ID located in the retailer ID extended memory table based on the FIID is written to the report.

Dormancy Report

The Dormancy Report displays the dormancy fees by institution, retailer ID, and card type. This report also provides information on the total dormancy fees that were assessed on stored value accounts contained with the Positive Balance File (PBF). The following is an example of a dormancy report

DORMANCY REPORT	NETWORK LLLL	DATE: DD-MM-YYYY	Page	1
TOTAL DORMANCY FEES PER INSTITUTION, RETAILER, AND CARD TYPE				
		TOTAL FEES		

INSTITUTION ID: BK01		\$ 0.00		
RETAILER ID: *****		\$ 0.00		
CARD TYPE: SR		\$ 0.00		
SN		\$ 0.00		
RETAILER ID: Retailer 1		\$ 0.00		
CARD TYPE: SR		\$ 0.00		
SN		\$ 0.00		
RETAILER ID: SPDH		\$ 0.00		
CARD TYPE: SR		\$ 0.00		
SN		\$ 0.00		
RETAILER ID: Retailer 2		\$ 0.00		
CARD TYPE: SR		\$ 0.00		
SN		\$ 0.00		
DORMANCY REPORT	NETWORK LLLL	DATE: DD-MMM-YYYY	Page	2
TOTAL DORMANCY FEES PER INSTITUTION, RETAILER, AND CARD TYPE				
		TOTAL FEES		

INSTITUTION ID: BK00		\$ 74.00		

INSTITUTION ID: BK00		\$ 74.00		
RETAILER ID: BANKTHREE		\$ 73.00		
CARD TYPE: SR		\$ 73.00		
SN		\$ 0.00		
RETAILER ID: SUPERONE		\$ 1.00		
CARD TYPE: SR		\$ 1.00		
SN		\$ 0.00		

Escheatment Report

The Escheatment Report displays dormancy fees by institution. The report also provides information about each account that is eligible for escheatment.

```

ESCHEATMENT REPORT      NETWORK  LLLL      DATE: DD-MMM-YYYY      Page  1
INSTITUTION              DETAIL INFORMATION FOR THIS INSTITUTION PER
ID      NAME              PRIMARY ACCOUNT NUMBER
BK01  BK01 - SV TEST BANK
  
```

```

          PRIOR TO DORMANCY FEE
PAN      BALANCE      DORMANCY FEE  BALANCE
CARD TYPE  DATE      LAST TRAN DATE  ACTIVATION STATE  RETAILER ID
  
```

```

ESCHEATMENT REPORT      NETWORK  LLLL      DATE: DD-MMM-YYYY      Page  2
INSTITUTION              TOTAL ESCHATEMENT AMOUNTS FOR EACH INSTITUTION'S
ID      NAME              STORED VALUE RETAILER ID
BK01  BK01 - SV TEST BANK
INSTITUTION TOTAL:      0.00
  
```

```

RETAILER ID      TOTAL
  
```

```

***** 0.00
KARIN3 0.00
SPDH    0.00
KARIN4 0.00
  
```

```

ESCHEATMENT REPORT      NETWORK  LLLL      DATE: DD-MMM-YYYY      Page  3
INSTITUTION              DETAIL INFORMATION FOR THIS INSTITUTION PER
ID      NAME              PRIMARY ACCOUNT NUMBER
BK00  BK00 - SV TEST BANK0
  
```

```

          PRIOR TO DORMANCY FEE
PAN      BALANCE      DORMANCY FEE  BALANCE
CARD TYPE  DATE      LAST TRAN DATE  ACTIVATION STATE  RETAILER ID
  
```

```

10990000000000001      900.00      24.00      876.00
SR      23-FEB-2004      03-JAN-2002  CA      BANKTHREE
10990000000000002      900.00      24.00      876.00
SR      23-FEB-2004      03-JAN-2002  CA      BANKTHREE
10990000000000003      900.00      24.00      876.00
SR      23-FEB-2004      03-JAN-2002  CA      BANKTHREE
10990000000000004      201.00      1.00      200.00
SR      11-JUN-2004      01-FEB-2002  CA      BANKTHREE
10990000000000005      100.00      1.00      99.00
SR      10-JUN-2004      10-FEB-2002  FL      SUPERONE
  
```

```

ESCHEATMENT REPORT      NETWORK  LLLL      DATE: DD-MMM-YYYY      Page  4
INSTITUTION              TOTAL ESCHATEMENT AMOUNTS FOR EACH INSTITUTION'S
ID      NAME              STORED VALUE RETAILER ID
BK00  BK00 - SV TEST BANK0
INSTITUTION TOTAL:      2927.00
  
```

```

RETAILER ID      TOTAL
  
```

```

SUPERONE      99.00
BANKTHREE     2828.00
  
```

ACI Worldwide Inc.

Index

A

- ACI Standard Device Configuration File (ACNF)
 - introduction, [2-2](#)
 - screen 8, [2-2](#)
- ACI Standard Device Response File (ARSP)
 - introduction, [2-4](#)
 - screen 5, [2-4](#)
- ACNF
 - see* ACI Standard Device Configuration File (ACNF)
- Additional card activation transaction request
 - message header and optional data fields, [4-9](#)
- Additional card activation transaction response
 - message header and optional data fields, [4-11](#)
- ARSP
 - see* ACI Standard Device Response File (ARSP)
- Authorization processing
 - adding CAF records, [3-3](#)
 - additional card activation transaction processing, [3-9](#)
 - CAF and PBF usages, [3-2](#)
 - CAF fields, [3-4](#)
 - card activation transaction processing, [3-6](#)
 - full redemption transaction processing, [3-12](#)
 - introduction, [3-1](#)
 - replenishment transaction processing, [3-10](#)

B

- Batch processing and reporting
 - introduction, [6-1](#)

C

- CAF
 - see* Cardholder Authorization File (CAF)
- CAF/PBF Cleanup (CARDCLN)
 - processing steps, [6-9](#)
- CAF/PBF Cleanup Program (CARDCLN)
 - introduction, [6-7](#)
 - reports, [6-8](#)
- Card activation transaction request
 - message header and optional data fields, [4-5](#)
- Card activation transaction response
 - message header and optional data fields, [4-7](#)
- Card Prefix File (CPF)
 - introduction, [2-6](#)
 - screen 6, [2-7](#)
 - screen 7, [2-8](#), [2-9](#)

- CARDCLN
 - see* CAF/PBF Cleanup (CARDCLN)
- Cardholder Authorization File (CAF)
 - introduction, [2-13](#)
 - screen 10, [2-13](#)
- CPF
 - see* Card Prefix File (CPF)

D

- Default account type, [1-3](#)

E

- ECF
 - see* Extract Configuration File (ECF)
- Extract Configuration File (ECF)
 - introduction, [2-14](#)
 - screen 7, [2-14](#)
- Extract Screen
 - introduction, [2-15](#)
 - screen 7, [2-15](#)
- Extract screen, [2-15](#)

F

- Full redemption transaction request
 - message header and optional data fields, [4-16](#)
- Full redemption transaction response
 - message header and optional data fields, [4-18](#)

H

- HFCLUP
 - see* History File Cleanup module (HFCLUP)
- History File Cleanup module (HFCLUP)
 - introduction, [6-2](#)
 - processing steps, [6-4](#)
 - program startup, [6-2](#)
 - reports, [6-3](#)

I

- IDF
 - see* Institution Definition File (IDF)
- Institution Definition File (IDF)
 - introduction, [2-16](#)
 - screen 1, [2-16](#)

ISO Host message formats
introduction, [5-1](#)
transaction codes, [5-2](#)

K

Key screen function keys, [2-22](#)

L

LCONF assigns and params
assigns, [1-5](#)
introduction, [1-5](#)
params, [1-6](#)

O

Original description field on ARSP screen 5, [2-5](#)

P

PBF
see Positive Balance File (PBF)
POS Transaction Log File (PTLF)
introduction, [2-17](#)
screen 4, [2-17](#)
Positive Balance File (File), [2-19](#)
PTLF
see POS Transaction Log File (PTLF)

R

Replenishment transaction request
message header and optional data fields, [4-12](#)
Replenishment transaction response
message header and optional data fields, [4-14](#)

S

SPDH message formats
introduction, [4-1](#)
transaction codes, [4-2](#)
Stored value card types
valid values, [1-4](#)
Stored Value Dormancy File (SVDF)
introduction, [2-49](#)
screen 1, [2-49](#)
Stored Value History File (SVHF)
card summary list screen 2, [2-31](#)
detail screen 3—format 1, [2-34](#)
detail screen 3—format 2, [2-40](#)
detail screen 3—format 3, [2-44](#)
detail screen 3—format 4, [2-48](#)
introduction, [2-20](#)
perusal selection screen 1, [2-23](#)
retailer summary list screen 2, [2-41](#)
retailer/terminal summary list screen 2, [2-45](#)
tran code summary list screen 2, [2-37](#)

Stored Value History File Extract module (SVHFX)
introduction, [6-6](#)

Stored value support, [1-2](#)

Stored value transaction types
additional card activation transaction, [1-2](#)
card activation transaction, [1-2](#)
full redemption transaction, [1-2](#)
replenishment transaction, [1-2](#)

SVDF
see Stored Value Dormancy File (SVDF)

SVHF
see Stored Value History File (SVHF)

SVHFX
see Stored Value History File Extract module (SVHFX)

T

Terminal messages
additional card activation transaction, [4-3](#)
card activation transaction, [4-3](#)
full redemption transaction, [4-3](#), [4-4](#)
replenishment transaction, [4-3](#)
Tokens, [1-8](#)

Index by Field Name

A

ACCOUNT BALANCE

- SVHF card summary list screen 2, [2-33](#), [2-39](#)
- SVHF detail screen 3, [2-36](#)

B

BEGIN DATE

- SVHF card summary list screen 2, [2-32](#), [2-38](#), [2-42](#), [2-46](#)
- SVHF key perusal selection screen 1, [2-24](#)
- SVHF perusal screen 1 — retailer perusal, [2-28](#)
- SVHF perusal screen 1 — retailer /terminal perusal, [2-29](#)
- SVHF perusal screen 1 — tran code perusal, [2-26](#)

C

CAF TEMPLATE LENGTH, [2-12](#)

CARD

- SVHF card summary list screen 2, [2-32](#), [2-38](#), [2-42](#), [2-46](#)
- SVHF key perusal selection screen 1, [2-25](#)
- SVHF perusal screen 1 — tran code perusal, [2-27](#)

CARD NUMBER

- SVHF detail screen 3, [2-35](#)

CARD TYPE

- SVDF screen, [2-50](#)

D

DAYS DORMANT

- SVDF screen, [2-51](#)

D/C IND

- SVHF card summary list screen 2, [2-33](#), [2-39](#)

DEBIT/CREDIT IND

- SVHF detail screen 3, [2-35](#)

E

END DATE

- SVHF card summary list screen 2, [2-32](#), [2-38](#), [2-42](#), [2-46](#)
- SVHF key perusal selection screen 1, [2-24](#)
- SVHF perusal screen 1 — retailer perusal, [2-28](#)
- SVHF perusal screen 1 — retailer/terminal perusal, [2-30](#)
- SVHF perusal screen 1 — tran code perusal, [2-27](#)

ESCHEATMENT, [2-11](#)

EXP DATE, [2-11](#)

EXP DURATION, [2-11](#)

F

FIID

- SVHF card summary list screen 2, [2-31](#), [2-37](#), [2-45](#)
- SVHF key perusal selection screen 1, [2-24](#)
- SVHF perusal screen 1 — retailer perusal, [2-28](#)
- SVHF perusal screen 1 — retailer/terminal perusal, [2-29](#)
- SVHF perusal screen 1 — tran code perusal, [2-25](#)

FLAT FEE

- SVDF screen, [2-50](#)

M

MAX BALANCE AS CASH, [2-10](#)

MAX BALANCE AS CASH REDEMPTION TRAN, [2-10](#)

MAX CARD BALANCE, [2-11](#)

MAX LOAD LIMIT, [2-10](#)

MBR NUM

- SVHF card summary list screen 2, [2-32](#), [2-38](#), [2-42](#), [2-46](#)

MEMBER NUMBER

- SVHF detail screen 3, [2-35](#)

MIN LOAD LIMIT, [2-10](#)

P

PERCENT FEE

- SVDF screen, [2-50](#)

PERIOD, [2-51](#)

R

REF NUM

- SVHF card summary list screen 2, [2-32](#), [2-38](#)
- SVHF detail screen 3, [2-36](#)

RESP CODE

- SVHF card summary list screen 2, [2-33](#), [2-39](#), [2-42](#), [2-47](#)
- SVHF detail screen 3, [2-36](#)

RETAILER ID, [2-11](#), [2-42](#), [2-46](#)
SVDF screen, [2-50](#)
SVHF detail screen 3, [2-34](#)
SVHF perusal screen 1 — retailer perusal, [2-28](#)
SVHF perusal screen 1 — retailer/terminal
perusal, [2-30](#)

S

ST
SVDF screen, [2-50](#)

T

TERM ID
SVHF perusal screen 1 — retailer/terminal
perusal, [2-30](#)

TERMINAL ID
SVHF detail screen 3, [2-34](#)

TRAN AMOUNT
SVHF card summary list screen 2, [2-33](#), [2-39](#)
SVHF detail screen 3, [2-35](#)

TRAN CODE, [2-38](#), [2-42](#), [2-46](#), [2-47](#)
SVHF card summary list screen 2, [2-32](#), [2-39](#), [2-42](#),
[2-47](#)
SVHF detail screen 3, [2-36](#)
SVHF key perusal selection screen 1 — tran code
perusal, [2-26](#)

TRAN DATE
SVHF card summary list screen 2, [2-33](#), [2-39](#), [2-42](#),
[2-47](#)
SVHF detail screen 3, [2-35](#)

TRAN TIME
SVHF card summary list screen 2, [2-33](#), [2-39](#), [2-43](#),
[2-47](#)
SVHF detail screen 3, [2-35](#)

TYP
SVHF card summary list screen 2, [2-31](#)
SVHF detail screen 3, [2-34](#)
SVHF key perusal selection screen 1, [2-24](#)
SVHF key perusal selection screen 1 — tran code
perusal, [2-25](#)
SVHF perusal screen 1 — retailer/terminal
perusal, [2-29](#)
SVHF perusal screen 1 — retailer perusal, [2-27](#)
SVHF retailer summary list screen 2, [2-41](#)
SVHF retailer/terminal summary list screen 2, [2-45](#)
SVHF tran code summary list screen 2, [2-37](#)

Index by Data Name

A

ALTKEY1.CARD-NUM, [2-35](#), [2-42](#)
ALTKEY1.CRD-NUM, [2-27](#)
ALTKEY1.FIID, [2-25](#)
ALTKEY1.TRAN-CDE, [2-26](#), [2-36](#), [2-38](#), [2-42](#), [2-47](#)
ALTKEY2.RETL-ID, [2-34](#), [2-42](#)
ALTKEY3.AMT, [2-33](#), [2-39](#)
ALTKEY3.BAL, [2-33](#), [2-39](#)
ALTKEY3.DB-CR-IND, [2-33](#), [2-39](#)
ALTKEY3.RETL-ID, [2-46](#), [2-50](#), [2-51](#)
ALTKEY3.TERM-ID, [2-34](#), [2-46](#)

D

DESCR.TBL, [2-5](#)
DESCR-TBL, [2-5](#)
DETL.AMT, [2-35](#)
DETL.BAL, [2-36](#)
DETL.DB-CR-IND, [2-35](#)
DETL.MBR-NUM, [2-38](#), [2-42](#), [2-46](#)
DETL.REF-NUM, [2-36](#), [2-38](#)
DETL.RESP-CDE, [2-33](#), [2-36](#), [2-39](#), [2-42](#), [2-47](#)

P

PRIKEY.CARD-NUM, [2-25](#), [2-32](#), [2-38](#), [2-46](#)
PRIKEY.FIID, [2-24](#), [2-31](#), [2-37](#), [2-45](#)
PRIKEY.TRAN-DAT-TIM, [2-24](#), [2-26](#), [2-27](#), [2-28](#),
[2-29](#), [2-30](#)

S

SVCPF.CAF-TPLT-LGTH, [2-12](#)
SVCPE.CRD-DURATION, [2-11](#)
SVCPE.CRD-EXP-DAT, [2-11](#)
SVCPE.MAX-AS-CASH, [2-10](#)
SVCPE.MAX-AS-CASH-REDEMPTION, [2-10](#)
SVCPE.MIN-LOAD, [2-10](#)
SVCPE-MAX-LOAD, [2-10](#)
SVDF.DAYS-DORMANT, [2-51](#)
SVDF.FLAT-FEE, [2-50](#)
SVDF.PR, [2-51](#)

SVDF.PRIKEY.ACT-ST, [2-50](#)
SVDF.PRIKEY.CRD-TYP, [2-50](#)
SVDF.PRIKEY.RETAILER-ID, [2-50](#)
SV-ESCHEATMENT-FLG, [2-11](#)
SV-MAX-CRD-BAL, [2-11](#)
SV-RETAILER-ID, [2-11](#)

ACI Worldwide Inc.